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(19) **United States**(12) **Patent Application Publication**
DONG et al.(10) **Pub. No.: US 2025/0277196 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **EPSPS MUTANTS AND METHOD OF ITS USES**(71) Applicant: **CROPEDIT BIOTECHNOLOGY INC., Beijing (CN)**(72) Inventors: **Yufeng DONG, Beijing (CN); Youlin QI, Beijing (CN); Yechun WU, Beijing (CN); Yujiao ZHOU, Beijing (CN); Xianggan LI, Beijing (CN); Yingying LI, Beijing (CN)**(21) Appl. No.: **18/263,514**(22) PCT Filed: **Jan. 28, 2021**(86) PCT No.: **PCT/CN2021/074029**

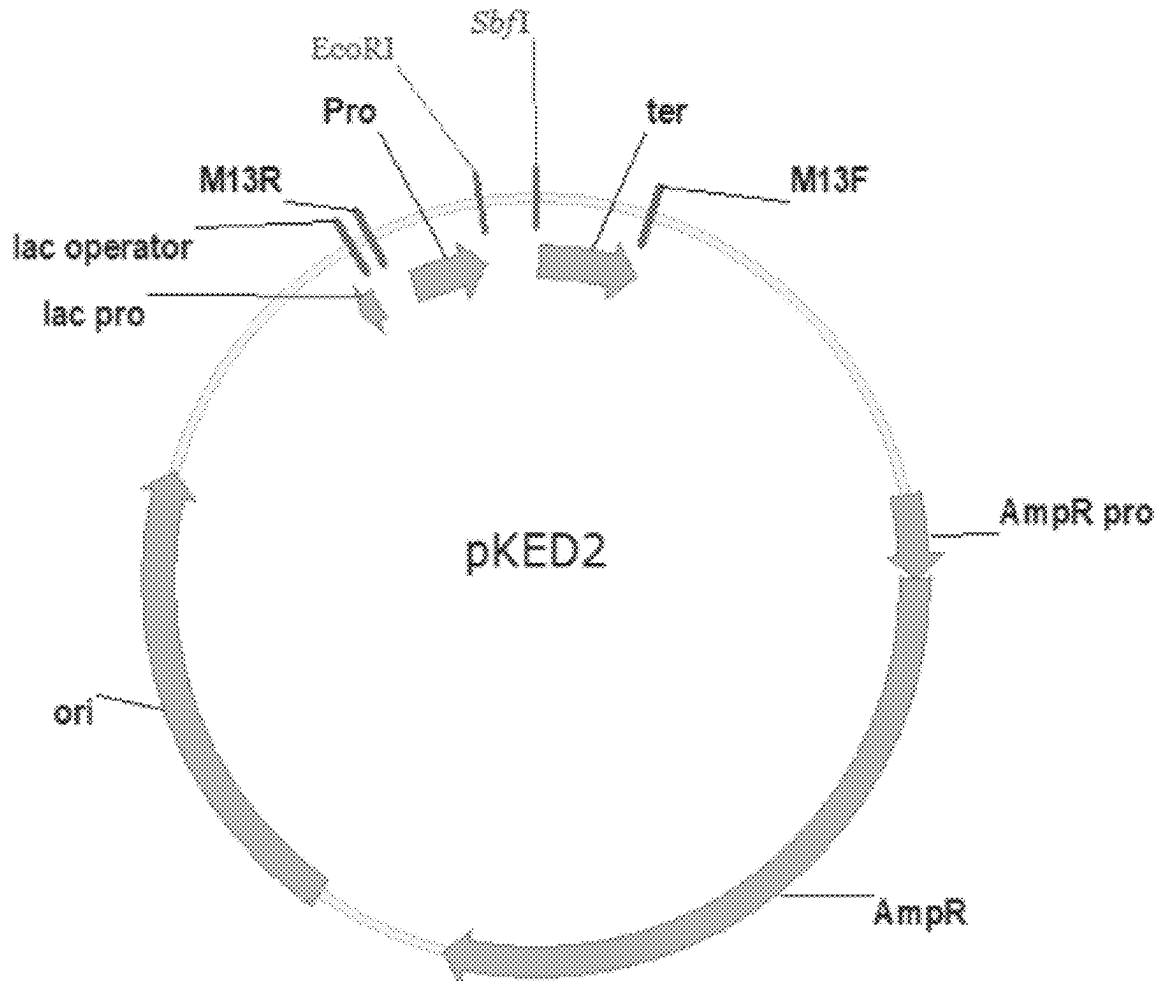
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(57)

ABSTRACT

The invention discloses gene sequences of a plant EPSPS mutant and its application in plants. As compared to the wild type, the present disclosure describes a plant EPSP synthase (EPSPS) polypeptide which comprises amino acid V mutating to amino acid A corresponding to the location at V403 or the analogous location at V403, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1.

Specification includes a Sequence Listing.

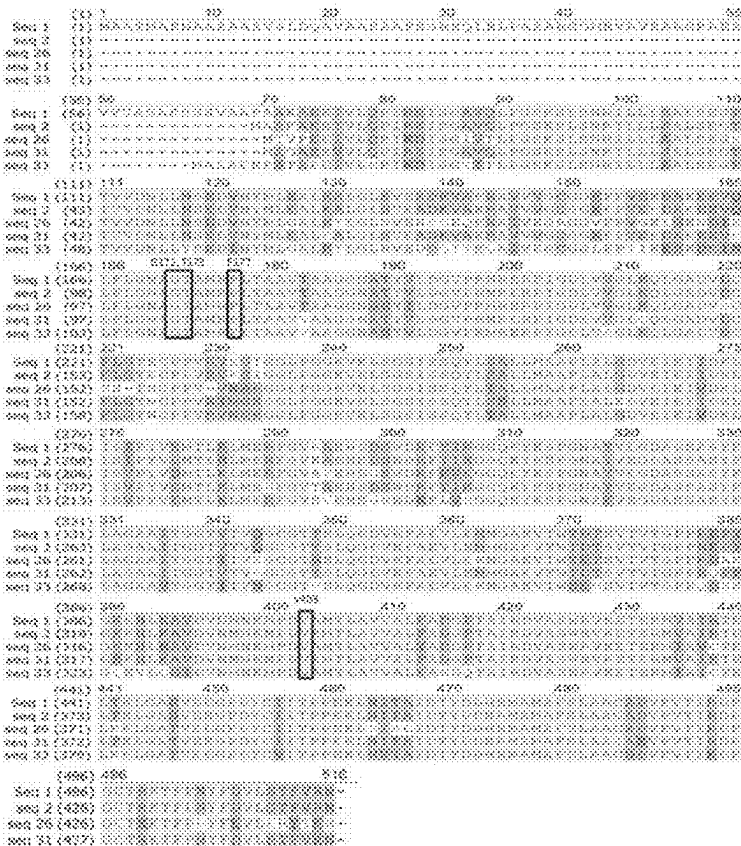


FIG. 1

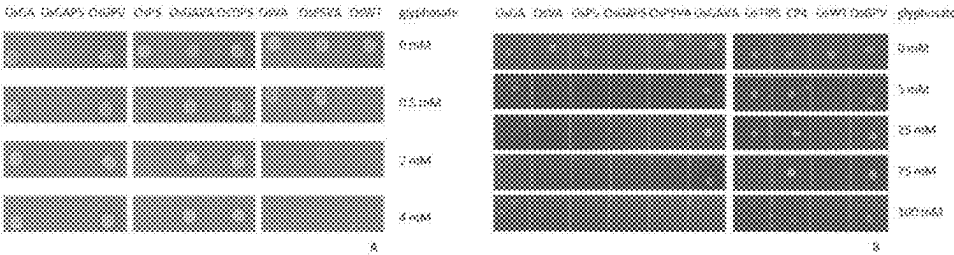


FIG. 2

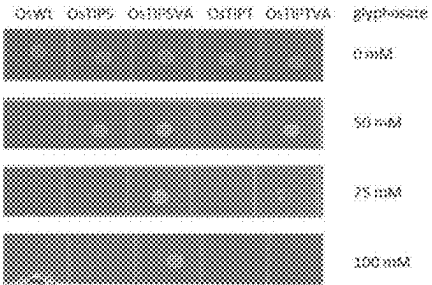


FIG. 3

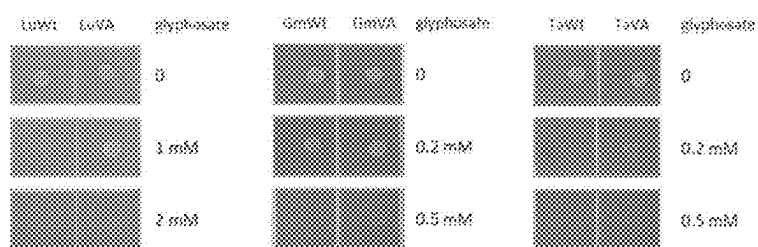


FIG. 4

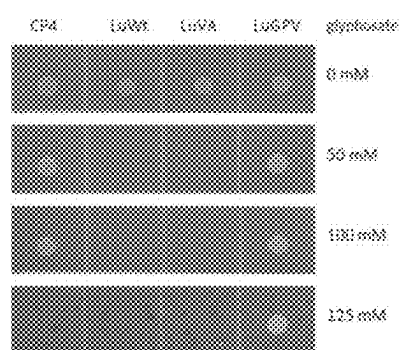


FIG. 5

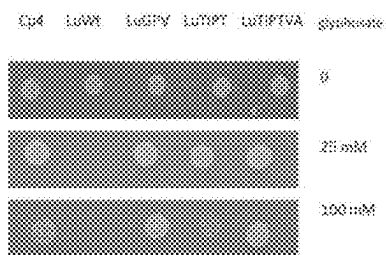


FIG. 6

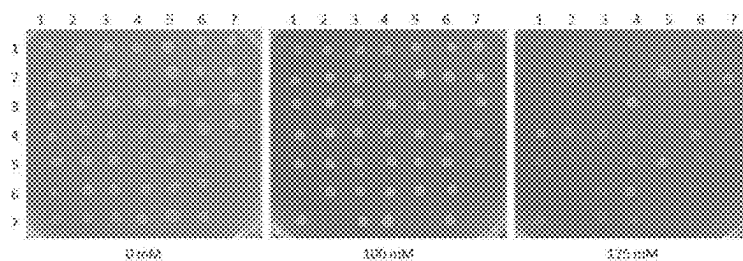


FIG. 7

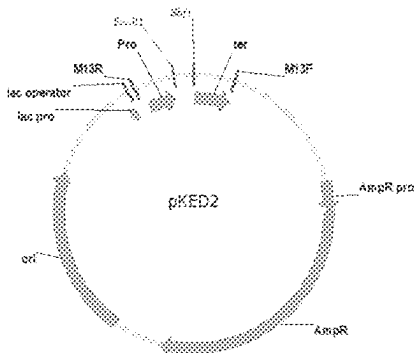


FIG. 8

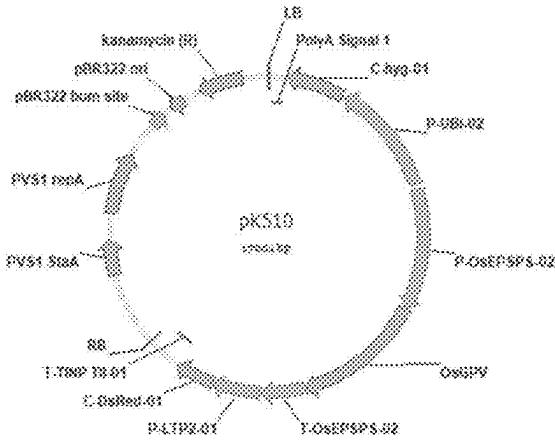


FIG. 9

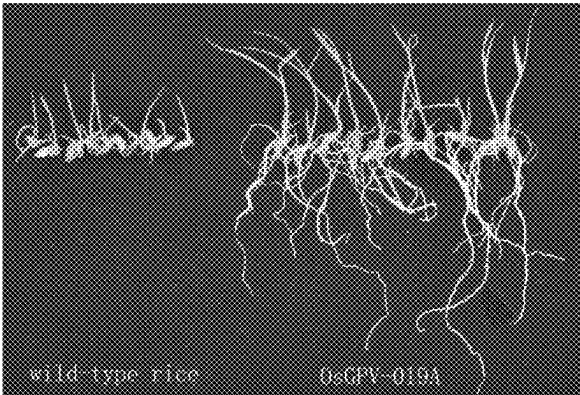


FIG. 10

EPSPS MUTANTS AND METHOD OF ITS USES

FIELD

[0001] The field of the invention relates to the field of molecular biology. More specifically, it pertains to amino acid mutations that confer tolerance to glyphosate and the use of it in the glyphosate tolerate plant.

BACKGROUND

[0002] The 5-enolpyruvylpyruvyl-3-phosphate synthase (EPSPS) encoded by the EPSPS gene is the sixth enzyme in the shikimate pathway. The enzyme catalyzes the conversion of phosphoenolpyruvate (PEP) and 3-phosphoshikimate (S3P) to phosphate and 5-enolpyruvate shikimate-3 phosphate (EPSP). The shikimic acid pathway is widely present in plants, bacteria and fungi, but no such pathway in animals. Glyphosate is one of the most sold chemical herbicides in the world. Its structure is a structural analogue of PEP. By competitively inhibiting the combination of EPSPS and PEP, it results in hindering the synthesis of downstream aromatic amino acids and the imbalance of downstream biosynthesis and metabolism, which effectively kills weeds. Glyphosate herbicide is almost lethal to all plants. In addition to EPSPS in plants, there are also EPSPS in some bacteria that are also sensitive to glyphosate. These glyphosate-sensitive EPSPS are classified as class I, the protein structure of the EPSPS in some microorganisms are similar to the protein structure of the EPSPS in plant, but the amino acid sequence similarity with class I EPSPS sequence is less than 50%, such as the CP4EPSPS, G2-aroA, . . . etc., this kind of EPSPS enzyme is insensitive to glyphosate and is classified as class II.

[0003] Glyphosate has become one of the most popular chemical herbicides because of its broad-spectrum, high-efficiency, low-toxicity, and low-cost. Its use in farmland can greatly save the cost of weeding and increase farmers' income. The resistance to glyphosate is one of the most popular traits among the plant varieties created by biotechnology. Up to now, there are about 7 genes in the about 221 approved transgenic events. Only two genes (mEPSPS and 2mEPSPS) are from plants, the other five genes are from microorganisms. As the public is concerned about the safety of introducing foreign genes into genetically modified crops, the product from the chemical mutagenesis and gene editing have become more accepted worldwide. In addition, Gene editing technology has become a new direction for the development of glyphosate-resistant crops because of its high efficiency, site-specific. With only fewer mutation to the gene of the plant, it is possible to obtain the target plant with ideal traits with high commercial value.

[0004] The present inventors discovered that certain amino acid mutation of the plant's original EPSPS enzyme can make the plant resistant to glyphosate without affecting the original enzyme function. The inventors also developed the application of these protein mutants and their encoded genes in transgenic plants, gene editing plants or other plant breeding, which can be used to cultivate plants with glyphosate resistance, especially crops, such as agricultural plants.

SUMMARY

[0005] The present disclosure relates to mutation position (s) of the amino acid(s) and the mutation type which comes

from the mutated EPSPS. Methods for generating glyphosate tolerant plants are also provided.

[0006] In one aspect, the mutated EPSPS polypeptides comprise the mutation V403A, wherein the amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0007] In one aspect, the mutated EPSPS polypeptides comprise the mutation G172A and V403A, wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1.

[0008] In one aspect, the mutated EPSPS polypeptides comprise the mutation P177S and V403A, wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1.

[0009] In another aspect, the mutated EPSPS polypeptides comprise the mutation G172A, P177S and/or V403A wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1.

[0010] In another aspect, the mutated EPSPS polypeptides comprise the mutation T173I, P177S and/or V403A wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1.

[0011] In another aspect, the mutated EPSPS polypeptides comprise the mutation T173I, P177T and/or V403A wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1.

[0012] Yet in another aspect, the mutated EPSPS polypeptides comprise the mutation V->A, wherein the position corresponds to the position of the original EPSPS amino acid sequence that is aligned to the position 403 of the SEQ ID NO:1.

[0013] Yet in another aspect, the mutated EPSPS polypeptides comprise the mutation G->A and V->A, wherein the positions correspond to the positions of the original EPSPS amino acid sequence that is aligned to the position 172 and 403 of the SEQ ID NO:1.

[0014] Yet in another aspect, the mutated EPSPS polypeptides comprise the mutation P->S and V->A, wherein the positions correspond to the positions of the original EPSPS amino acid sequence that is aligned to the position 177 and 403 of the SEQ ID NO:1.

[0015] Yet in another aspect, the mutated EPSPS polypeptides comprise the mutation G->A, P->S and/or V->A, wherein the positions correspond to the positions of the original EPSPS amino acid sequence that is aligned to the position 172, 177 and/or 403 of the SEQ ID NO:1.

[0016] Yet in another aspect, the mutated EPSPS polypeptides comprise the mutation T->I, P->S and/or V->A, wherein the positions correspond to the positions of the original EPSPS amino acid sequence that is aligned to the position 173, 177 and/or 403 of the SEQ ID NO:1.

[0017] Yet in another aspect, the mutated EPSPS polypeptides comprise the mutation T->I, P->T and/or V->A, wherein the positions correspond to the positions of the original EPSPS amino acid sequence that is aligned to the position 173, 177 and/or 403 of the SEQ ID NO:1.

[0018] In yet another aspect, the mutated EPSPS polypeptides comprise the mutation G172A, P177S and/or V403A and at least one or more amino acid mutations selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D,

(22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0019] In yet another aspect, the mutated EPSPS polypeptides comprise the mutation G172A, P177S and/or V403A and at least one or more analogous amino acid mutations selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO: 1.

[0020] In other embodiments, the mutated EPSPS polypeptide comprises the mutation A70V, G172A, P177S, V403A and/or R514S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0021] In other embodiments, the mutated EPSPS polypeptide comprises the mutation A73T, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0022] In other embodiments, the mutated EPSPS polypeptide comprises the mutation A73T, E86G, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0023] In other embodiments, the mutated EPSPS polypeptide comprises the mutation A145G, G172A, P177S, V403A, T442S, and/or D472E, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0024] In other embodiments, the mutated EPSPS polypeptide comprises the mutation G172A, P177S, V403A and/or F502V, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0025] In other embodiments, the mutated EPSPS polypeptide comprises the mutation G172A, P177S, Y252F and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0026] In other embodiments, the mutated EPSPS polypeptide comprises the mutation E137D, G172A, P177S, L253M and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0027] In other embodiments, the mutated EPSPS polypeptide comprises the mutation G172A, P177S, K399N, V403A and/or T466S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0028] In other embodiments, the mutated EPSPS polypeptide comprises the mutation A73G, L134F, G172A, P177S, G265V, S374R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0029] In other embodiments, the mutated EPSPS polypeptide comprises the mutation A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0030] In other embodiments, the mutated EPSPS polypeptide comprises the mutation E163K, G172A, P177S, A216V, L253M, H389R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0031] In other embodiments, the mutated EPSPS polypeptide comprises the mutation G172A, P177S, N246D and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0032] In other embodiments, the mutated EPSPS polypeptide comprises the mutation G172A, M175A, P177S, M258L, K356E, V403A and/or F512I, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0033] In other embodiments, the mutated EPSPS polypeptide comprises the mutation A73S, G172A, P177S, L214F and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0034] In other embodiments, the mutated EPSPS polypeptide comprises the mutation G172A, P177S, V208A, L264H, V403A and/or R514K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0035] In other embodiments, the mutated EPSPS polypeptide comprises the mutation V113M, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0036] Also provided are recombinant DNA constructs comprising the polynucleotides that encode the polypeptide disclosed herein; plant cells comprising a polynucleotide disclosed herein or a recombinant DNA construct comprising such. In some embodiments, the plant cell is a rice cell. In some embodiments, the plant is rice.

[0037] In an aspect, methods of generating glyphosate tolerant plants are provided herein. The method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0038] In an aspect, methods of generating glyphosate tolerant plants are provided herein. The method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A and V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0039] In an aspect, methods of generating glyphosate tolerant plants are provided herein. The method comprises expressing in a regenerable plant cell a recombinant DNA

construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises P177S and V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0040] In some embodiments, the method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0041] In some embodiments, the method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises T173I, P177S and/or V403A wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0042] In some embodiments, the method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises T173I, P177T and/or V403A wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0043] In some embodiments, the method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A, P177S and/or V403A and at least one amino acid mutation selected from the group consisting of (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0044] In an aspect, methods of generating glyphosate tolerant plants are provided herein. The method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to V403A, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0045] In an aspect, methods of generating glyphosate tolerant plants are provided herein. The method comprises expressing in a regenerable plant cell a recombinant DNA

construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to G172A and an amino acid mutation that is analogous to V403A, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0046] In an aspect, methods of generating glyphosate tolerant plants are provided herein. The method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0047] In some embodiments, the method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to G172A, an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0048] In some embodiments, the method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to T173I, an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0049] In some embodiments, the method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to T173I, an amino acid mutation that is analogous to P177T and/or an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0050] In some embodiments, the method comprises expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to G172A, an amino acid mutation that is analogous to P177S, and/or an amino acid

mutation that is analogous to V403A and at least one amino acid mutation that is analogous to the amino acid mutation selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0051] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises A70V, G172A, P177S, V403 and/or R514S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0052] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises A73T, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct. In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises A73T, E86G, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0053] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises A145G, G172A, P177S, V403A, T442S, and/or D472E, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0054] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises G172A, P177S, V403A and/or F502V, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0055] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises G172A, P177S, Y252F and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0056] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises E137D, G172A, P177S, L253M and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0057] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises G172A, P177S, K399N, V403A and/or T466S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0058] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises A73G, L134G, G172A, P177S, G265V, S374R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0059] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0060] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0061] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises G172A, P177S, N247D and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0062] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0063] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises A73S, G172A, P177S, L214F

and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0064] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises G172A, P177S, V208A, L264H, V403A and/or R514K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0065] In other embodiments, the method comprises expressing in a plant cell a recombinant DNA construct comprising a polynucleotide that encodes a plant EPSPS polypeptide that comprises V113M, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1, and generating a glyphosate tolerant plant that comprises in its genome the recombinant DNA construct.

[0066] In an aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises V403A, wherein the amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0067] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises G172A and V403A wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0068] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises P177S and V403A wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0069] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, P177S and/or V403A, wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0070] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises T173I, P177S and/or V403A, wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0071] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises

T173I, P177T and/or V403A, wherein the amino acid mutation positions correspond to the amino acid positions set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0072] Yet in another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, P177S and/or V403A, and at least one amino acid selecting from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0073] In an aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to V403A, wherein the amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0074] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to G172A and an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0075] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to P177S and an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0076] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to G172A, an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0077] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to T173I, an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous

amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0078] In another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to T173I, an amino acid mutation that is analogous to P177T and/or an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0079] Yet in another aspect, methods of generating glyphosate tolerant plants are provided herein, in which an endogenous plant EPSPS gene (in a plant cell) is modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to G172A, an amino acid mutation that is analogous to P177S, and/or an amino acid mutation that is analogous to V403A and at least one amino acid mutation that is analogous to one amino acid mutation selecting from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1; and a glyphosate tolerant plant is grown from the plant cell.

[0080] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises A70V, G172A, P177S, V403 and/or R514S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0081] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises A73T, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0082] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises A73T, E86G, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0083] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises A145G, G172A, P177S, V403A, T442S, and/or D472E, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0084] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises G172A, P177S, V403A and/or F502V, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0085] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises G172A, P177S, Y252F and/or

V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0086] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises E137D, G172A, P177S, L253M and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0087] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises G172A, P177S, K399N, V403A and/or T466S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0088] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises A73G, L134G, G172A, P177S, G265V, S374R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0089] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0090] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0091] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises G172A, P177S, N247D and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0092] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0093] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises A73S, G172A, P177S, L214F and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0094] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises G172A, P177S, V208A, L264H, V403A and/or R514K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0095] In other embodiments, the modified endogenous plant EPSPS gene encodes a glyphosate tolerant EPSPS protein that comprises V113M, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0096] The endogenous plant EPSPS gene can be modified by any methods known to one skilled in the art, including, but not limited to: CRISPR/Cas guide RNA-mediated system, Zn-finger nuclease-mediated system, meganuclease-mediated system, and/or an oligonucleobase-mediated system.

[0097] Polynucleotide constructs that provide a guide RNA in a plant cell are provided herein in which the guide RNA targets an endogenous EPSPS gene of the plant cell and the polynucleotide construct further comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene that encodes a plant EPSPS polypeptide comprising V403A, wherein the amino acid mutation position corresponds to the amino acid set forth in SEQ ID NO:1.

[0098] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A and V403A, wherein the amino acid position corresponds to the amino acid mutation position set forth in SEQ ID NO:1.

[0099] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises P177S and V403A, wherein the amino acid position corresponds to the amino acid mutation position set forth in SEQ ID NO:1.

[0100] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid mutation position set forth in SEQ ID NO:1.

[0101] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises T173I, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid mutation position set forth in SEQ ID NO:1.

[0102] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises T173I, P177T and/or V403A, wherein the amino acid position corresponds to the amino acid mutation position set forth in SEQ ID NO:1.

[0103] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A, P177S, and/or V403A, and at least one or more amino acid mutation selecting from the group consisting of (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid position corresponds to the amino acid mutation position set forth in SEQ ID NO:1.

[0104] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to V403A, wherein the amino acid position corresponds to the analogous amino acid mutation position set forth in SEQ ID NO:1.

[0105] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to G172A and an amino acid mutation that is analogous to V403A, wherein each amino acid position corresponds to the amino acid mutation position set forth in SEQ ID NO:1.

[0106] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to P177S and an amino acid mutation that is analogous to V403A, wherein the amino acid position corresponds to the analogous amino acid mutation position set forth in SEQ ID NO:1.

[0107] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to G172A, an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A, wherein each amino acid position corresponds to the analogous amino acid mutation position set forth in SEQ ID NO:1.

[0108] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to T173I, an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A, wherein each amino acid position corresponds to the analogous amino acid mutation position set forth in SEQ ID NO:1.

[0109] In some embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises an amino acid mutation that is analogous to T173I, an amino acid mutation that is analogous to P177T and/or an amino acid mutation that is analogous to V403A, wherein each amino acid position corresponds to the analogous amino acid mutation position set forth in SEQ ID NO:1.

[0110] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises A70V, G172A, P177S, V403 and/or R514S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0111] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises A73T, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0112] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises A73T, E86G, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0113] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises A145G, G172A, P177S, V403A, T442S, and/or D472E, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0114] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A, P177S, V403A and/or F502V, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0115] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A, P177S, Y252F and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0116] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises E137D, G172A, P177S, L253M and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0117] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A, P177S, K399N, V403A and/or T466S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0118] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises A73G, L134G, G172A, P177S, G265V, S374R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0119] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0120] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0121] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A, P177S, N247D and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0122] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0123] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises A73S, G172A, P177S, L214F and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0124] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises G172A, P177S, V208A, L264H, V403A and/or R514K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0125] In other embodiments, the polynucleotide construct comprises one or more polynucleotide modification templates to generate a modified endogenous EPSPS gene encoding a plant EPSPS polypeptide that comprises V113M, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0126] Also provided herein are glyphosate tolerant plants that express an endogenous EPSPS polypeptide that has V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0127] Also provided herein are glyphosate tolerant plants that express an endogenous EPSPS polypeptide that has G172A and V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0128] Also provided herein are glyphosate tolerant plants that express an endogenous EPSPS polypeptide that has P177S and V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0129] Also provide herein are glyphosate tolerant plants that express an endogenous EPSPS polypeptide that has G172A, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0130] Also provide herein are glyphosate tolerant plants that express an endogenous EPSPS polypeptide that has T173I, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0131] Also provide herein are glyphosate tolerant plants that express an endogenous EPSPS polypeptide that has T173I, P177T and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0132] Also provide herein are glyphosate tolerant plants that express an endogenous EPSPS polypeptide that has

G172A, P177S and/or V403A, and one or more amino acid mutations selecting from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0133] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:3.

[0134] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A and V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:4.

[0135] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has P177S and V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:5.

[0136] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:6.

[0137] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has T173I, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:37.

[0138] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has T173I, P177T and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:38.

[0139] Also provide herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A, P177S and/or V403A, and one or more amino acid mutations selecting from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein

each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0140] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has A70V, G172A, P177S, V403 and/or R514S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:10.

[0141] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has A73T, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:11.

[0142] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has A73T, E86G, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:12.

[0143] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has A145G, G172A, P177S, V403A, T442S, and/or D472E, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:13.

[0144] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A, P177S, V403A and/or F502V, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:14.

[0145] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A, P177S, Y252F and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:15.

[0146] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has E137D, G172A, P177S, L253M and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:16.

[0147] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A, P177S, K399N, V403A and/or T466S, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:17.

[0148] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has A73G, L134G, G172A, P177S, G265V, S374R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:

1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:18.

[0149] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:19.

[0150] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has E163K, G172A, P177S, A216V, L253M, H389R and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:20.

[0151] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A, P177S, N247D and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:21.

[0152] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A, M175A, P177S, M258L, K356E, V403A and/or F512I, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:22.

[0153] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has A73S, G172A, P177S, L214F and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:23.

[0154] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has G172A, P177S, V208A, L264H, V403A and/or R514K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:24.

[0155] Also provided herein are glyphosate tolerant rice plants that express an endogenous EPSPS polypeptide that has V113M, G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant rice plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:25.

[0156] Also provided herein are glyphosate tolerant flax plants that express an endogenous EPSPS polypeptide that has V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant flax plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:27.

[0157] Also provided herein are glyphosate tolerant flax plants that express an endogenous EPSPS polypeptide that has T173I and P177T, wherein the amino acid positions

correspond to the amino acid positions set forth in SEQ ID NO:1. A glyphosate tolerant flax plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:28.

[0158] Also provided herein are glyphosate tolerant flax plants that express an endogenous EPSPS polypeptide that has T173I, P177T and/or V403A, wherein each amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant flax plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:29.

[0159] Also provide herein are glyphosate tolerant flax plants that express an endogenous EPSPS polypeptide that has G172A, P177S and/or V403A, wherein each amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant flax plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:30.

[0160] Also provide herein are glyphosate tolerant flax plants that express an endogenous EPSPS polypeptide that has G172A, P177S and/or V403A, and one or more analogous amino acid mutations selecting from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1.

[0161] Also provided herein are glyphosate tolerant wheat plants that express an endogenous EPSPS polypeptide that has V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1. A glyphosate tolerant wheat plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:32.

[0162] Also provide herein are glyphosate tolerant wheat plants that express an endogenous EPSPS polypeptide that has G172A, P177S, and/or V403A, and one or more analogous amino acid mutations selecting from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1.

[0163] Also provided herein are glyphosate tolerant soybean plants that express an endogenous EPSPS polypeptide that has V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. A glyphosate tolerant soybean plant may express a plant EPSPS polypeptide having the sequence set forth in SEQ ID NO:34.

[0164] Also provide herein are glyphosate tolerant soybean plants that express an endogenous EPSPS polypeptide that has G172A, P177S and/or V403A, and one or more analogous amino acid mutations selecting from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4)

D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1. Also provided is a method of producing a glyphosate tolerant plant. The method includes introducing into plant cells a recombinant oligonucleotide with a mutant EPSPS gene that expresses a EPSPS protein that is mutated at amino acid position V403 in a rice EPSPS protein (SEQ ID NO: 1) or at an analogous amino acid residue in an EPSPS homolog, wherein V403 is changed to A; selecting a plant cell exhibiting improved tolerance to glyphosate as compared to a corresponding wild-type plant cell; and regenerating a herbicide resistant or tolerant plant having a mutated EPSPS gene from said selected plant cell.

[0165] Also provided is a method of producing a glyphosate tolerant plant. The method includes introducing into plant cells a recombinant oligonucleotide with a mutant EPSPS gene that expresses a EPSPS protein that is mutated at amino acid position G172 and V403 in a rice EPSPS protein (SEQ ID NO: 1) or at an analogous amino acid residue in an EPSPS homolog, wherein G172 is changed to A and V403 is changed to A; selecting a plant cell exhibiting improved tolerance to glyphosate as compared to a corresponding wild-type plant cell; and regenerating a herbicide resistant or tolerant plant having a mutated EPSPS gene from said selected plant cell.

[0166] Also provided is a method of producing a glyphosate tolerant plant. The method includes introducing into plant cells a recombinant oligonucleotide with a mutant EPSPS gene that expresses a EPSPS protein that is mutated at amino acid position P177 and V403 in a rice EPSPS protein (SEQ ID NO: 1) or at an analogous amino acid residue in an EPSPS homolog, wherein P177 is changed to S and V403 is changed to A; selecting a plant cell exhibiting improved tolerance to glyphosate as compared to a corresponding wild-type plant cell; and regenerating a herbicide resistant or tolerant plant having a mutated EPSPS gene from said selected plant cell.

[0167] Also provided is a method of producing a glyphosate tolerant plant. The method includes introducing into plant cells a recombinant oligonucleotide with a mutant EPSPS gene that expresses a EPSPS protein that is mutated at amino acid position G172, P177 and/or V403 in a rice EPSPS protein (SEQ ID NO: 1) or at an analogous amino acid residue in an EPSPS homolog, wherein G172 is changed to A, P177 is changed to S and/or V403 is changed to A; selecting a plant cell exhibiting improved tolerance to glyphosate as compared to a corresponding wild-type plant cell; and regenerating a herbicide resistant or tolerant plant having a mutated EPSPS gene from said selected plant cell.

[0168] Also provided is a method of producing a glyphosate tolerant plant. The method includes introducing into plant cells a recombinant oligonucleotide with a mutant EPSPS gene that expresses a EPSPS protein that is mutated at amino acid position T173, P177 and/or V403 in a rice EPSPS protein (SEQ ID NO: 1) or at an analogous amino acid residue in an EPSPS homolog, wherein T173 is changed to I, P177 is changed to S and/or V403 is changed to A; selecting a plant cell exhibiting improved tolerance to

glyphosate as compared to a corresponding wild-type plant cell; and regenerating a herbicide resistant or tolerant plant having a mutated EPSPS gene from said selected plant cell. **[0169]** Also provided is a method of producing a glyphosate tolerant plant. The method includes introducing into plant cells a recombinant oligonucleotide with a mutant EPSPS gene that expresses a EPSPS protein that is mutated at amino acid position T173, P177 and/or V403 in a rice EPSPS protein (SEQ ID NO: 1) or at an analogous amino acid residue in an EPSPS homolog, wherein T173 is changed to I, P177 is changed to T and/or V403 is changed to A; selecting a plant cell exhibiting improved tolerance to glyphosate as compared to a corresponding wild-type plant cell; and regenerating a non-transgenic herbicide resistant or tolerant plant having a mutated EPSPS gene from said selected plant cell.

BRIEF DESCRIPTION OF THE DRAWINGS

[0170] FIG. 1 shows a multiple alignment among SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 26, SEQ ID NO: 31 and SEQ ID NO: 33.

[0171] FIG. 2 shows growth of *E. coli* cells expressing rice EPSPS protein with G172A, P177S and/or V403A mutants and its combination mutants in different glyphosate concentration media.

[0172] FIG. 3 shows growth of *E. coli* cells expressing rice EPSPS protein with T173I, P177S/T and/or V403A mutants and its combination mutants in different glyphosate concentration media.

[0173] FIG. 4 shows the growth status of *E. coli* cells expressing flax or wheat or soybean EPSPS protein with V403A mutants in different glyphosate concentration media.

[0174] FIG. 5 shows the growth status comparison between *E. coli* cells expressing flax EPSPS protein with G172A, P177S and V403A mutants and *E. coli* cells expressing CP4 EPSPS protein in different glyphosate concentration media.

[0175] FIG. 6 shows the growth status of *E. coli* cells expressing flax EPSPS protein with G172A, T173I, P177T/S and/or V403A mutants in different glyphosate concentration media.

[0176] FIG. 7 shows the glyphosate resistance comparison of *E. coli* cells expressing rice EPSPS protein with multiple mutants at a concentration of 125 ml glyphosate herbicide.

[0177] FIG. 8 shows the construct pKED2 vector.

[0178] FIG. 9 shows the pk510 vector which comprises one embodiment of the present disclosure.

[0179] FIG. 10 shows comparison between wild-type rice plants and transgenic rice plants and their ability to be glyphosate resistant.

DETAILED DESCRIPTION

[0180] The term “EPSPS homolog” or any variation thereof refers to an EPSPS gene or EPSPS gene product found in another plant species that performs the same or substantially the same biological function as the EPSPS genes disclosed herein and where the nucleic acid sequences or polypeptide sequences (of the EPSPS gene product) are said to be “identical” or at least 50% similar (also referred to as “percent identity” or “substantially identical”) as described below. Two polynucleotides or polypeptides are identical if the sequence of nucleotides or amino acid residues, respectively, in the two sequences is the same when aligned for

maximum correspondence as described below. The terms “identical” or “percent identity,” in the context of two or more nucleic acids or polypeptide sequences, refer to two or more sequences or subsequences that are the same or have a specified percentage of amino acid residues or nucleotides that are the same, when compared and aligned for maximum correspondence over a comparison window, as measured using one of the following sequence comparison algorithms or by manual alignment and visual inspection. For polypeptides where sequences differ in conservative substitutions, the percent sequence identity may be adjusted upwards to correct for the conservative nature of the substitution. Means for making this adjustment are well known to those of skill in the art. Typically this involves scoring a conservative substitution as a partial rather than a full mismatch, thereby increasing the percentage sequence identity. Thus, for example, where an identical amino acid is given a score of 1 and a non-conservative substitution is given a ‘score of zero, a conservative substitution is given a score between zero and 1. The scoring of conservative substitutions is calculated according to, e.g., the algorithm of Meyers & Miller, *Computer Applic. Biol. Sci.* 4: 11-17 (1988) e.g., as implemented in the program PC/GENE (Intelligenetics, Mountain View, Calif, USA).

[0181] The phrases “substantially identical,” and “percent identity” in the context of two nucleic acids or polypeptides, refer to sequences or subsequences that have at least 50%, advantageously 60%, preferably 70%, more preferably 80%, and most preferably 90-95% nucleotide or amino acid residue identity when aligned for maximum correspondence over a comparison window as measured using one of the following sequence comparison algorithms or by manual alignment and visual inspection. This definition also refers to the complement of a test sequence, which has substantial sequence or subsequence complementarity when the test sequence has substantial identity to a reference sequence.

[0182] “Operably linked” refers to a juxtaposition of genetic elements, wherein the elements are in a relationship permitting them to operate in the expected manner. For instance, a 5' regulatory sequence is operably linked to a coding sequence if the 5' regulatory sequence functions to initiate transcription of the coding sequence.

[0183] As used herein, a “glyphosate-resistant” cell or plant refers to a cell or plant that can survive or continue to grow in the presence of certain concentrations of glyphosate that typically kill or inhibit the growth of other cells or plants. Growth includes, for instance, photosynthesis, increased rooting, increased height, increased mass, or development of new leaves. In one embodiment, a glyphosate-resistant cell can grow and divide on a culture medium containing 50 mg/l or more glyphosate. Preferably, a glyphosate-resistant cell can grow and divide on a culture medium containing 100 mg/l or more glyphosate, such as 200 mg/l, 300 mg/l or 400 mg/l glyphosate. More preferably, a glyphosate-resistant cell can grow and divide on a culture medium containing 500 mg/l or more glyphosate, such as 600 mg/l. For purposes of the present invention, the term “glyphosate” includes any herbicidally effective form of N-phosphonomethylglycine (including any salt thereof) and other forms which result in the production of the glyphosate anion in plants.

Sequence Listing Details

[0184] SEQ ID NO:1 is the full length amino acid sequence of a wild type rice EPSPS protein (GenBank entry XP_015643046.1), wherein its N-terminal comprises a chloroplast transit peptide. SEQ ID NO:1 is to be used herein as a reference EPSPS sequence and all mutation positions are marked according to SEQ ID NO:1.

[0185] SEQ ID NO:2 is the amino acid sequence of a wild type rice EPSPS protein. Comparing to SEQ ID NO: 1, the chloroplast transit peptide is removed from SEQ ID NO:2.

[0186] SEQ ID NO:3 is the mutated version of the EPSPS sequence from rice that contains V403A mutations.

[0187] SEQ ID NO:4 is the mutated version of the EPSPS sequence from rice that contains G172A and V403A mutations.

[0188] SEQ ID NO:5 is the mutated version of the EPSPS sequence from rice that contains P177S and V403A mutations.

[0189] SEQ ID NO:6 is the mutated version of the EPSPS sequence from rice that contains G172A, P177S and V403A mutations.

[0190] SEQ ID NO:7 is the mutated version of the EPSPS sequence from rice that contains G172A mutations.

[0191] SEQ ID NO:8 is the mutated version of the EPSPS sequence from rice that contains P177S mutations.

[0192] SEQ ID NO:9 is the mutated version of the EPSPS sequence from rice that contains G172A and P177S mutations.

[0193] SEQ ID NO:10 is the mutated version of the EPSPS sequence from rice that contains A70V, G172A, P177S, V403A and R514S mutations.

[0194] SEQ ID NO:11 is the mutated version of the EPSPS sequence from rice that contains A73T, G172A, P177S and V403A mutations.

[0195] SEQ ID NO:12 is the mutated version of the EPSPS sequence from rice that contains A73T, E86G, G172A, P177S and V403A mutations.

[0196] SEQ ID NO:13 is the mutated version of the EPSPS sequence from rice that contains A145G, G172A, P177S, V403A, T442S and D472E mutations.

[0197] SEQ ID NO:14 is the mutated version of the EPSPS sequence from rice that contains G172A, P177S, V403A and F502V mutations.

[0198] SEQ ID NO:15 is the mutated version of the EPSPS sequence from rice that contains G172A, P177S, Y252F and V403A mutations.

[0199] SEQ ID NO:16 is the mutated version of the EPSPS sequence from rice that contains E137D, G172A, P177S, L253M and V403A mutations.

[0200] SEQ ID NO:17 is the mutated version of the EPSPS sequence from rice that contains G172A, P177S, K399N, V403A and T466S mutations.

[0201] SEQ ID NO:18 is the mutated version of the EPSPS sequence from rice that contains A73G, L134F, G172A, P177S, G265V, S374R and V403A mutations.

[0202] SEQ ID NO:19 is the mutated version of the EPSPS sequence from rice that contains A73K, D114H, V136M, K161E, G172A, P177S, A260T and V403A mutations.

[0203] SEQ ID NO:20 is the mutated version of the EPSPS sequence from rice that contains E163K, G172A, P177S, A216V, L253M, H389R and V403A mutations.

[0204] SEQ ID NO:21 is the mutated version of the EPSPS sequence from rice that contains G172A, P177S, N246D and V403A mutations.

[0205] SEQ ID NO:22 is the mutated version of the EPSPS sequence from rice that contains G172A, M175A, P177S, M258L, K356E, V403A and F512I mutations.

[0206] SEQ ID NO:23 is the mutated version of the EPSPS sequence from rice that contains A73S, G172A, P177S, L214F and V403A mutations.

[0207] SEQ ID NO:24 is the mutated version of the EPSPS sequence from rice that contains G172A, P177S, V208A, L264H, V403A and R514K mutations.

[0208] SEQ ID NO:25 is the mutated version of the EPSPS sequence from rice that contains V113M, G172A, P177S and V403A mutations.

[0209] SEQ ID NO:26 is the amino acid sequence of a wild type flax EPSPS2 translated from Lus10000788 (Sauer N J, Narváez-Vásquez, Javier, Mozoruk J, et al. Oligonucleotide-Mediated Genome Editing Provides Precision and Function to Engineered Nucleases and Antibiotics in Plants [J]. Plant Physiology, 2016:1917-1928.), wherein its chloroplast transit peptide is removed from SEQ ID NO:26.

[0210] SEQ ID NO:27 is the mutated version of the EPSPS sequence from flax that contains V403A mutations.

[0211] SEQ ID NO:28 is the mutated version of the EPSPS sequence from flax that contains T173I and P177T mutations.

[0212] SEQ ID NO:29 is the mutated version of the EPSPS sequence from flax that contains T173I, P177T and V403A mutations.

[0213] SEQ ID NO:30 is the mutated version of the EPSPS sequence from flax that contains G172A, P177S and V403A mutations.

[0214] SEQ ID NO:31 is the amino acid sequence of a wild type wheat EPSPS protein presented as GenBank entry ALK27163.1, wherein its chloroplast transit peptide is removed from SEQ ID NO:31.

[0215] SEQ ID NO:32 is the mutated version of the EPSPS sequence from wheat that contains V403A mutations.

[0216] SEQ ID NO:33 is the amino acid sequence of a wild type soybean EPSPS protein presented as GenBank entry XP_003521857.1, wherein its chloroplast transit peptide is removed from SEQ ID NO:33.

[0217] SEQ ID NO:34 is the mutated version of the EPSPS sequence from soybean that contains V403A mutations.

[0218] SEQ ID NO:35 is the mutated version of the EPSPS sequence from rice that contains T173I and P177S mutations.

[0219] SEQ ID NO:36 is the mutated version of the EPSPS sequence from rice that contains T173I and P177T mutations.

[0220] SEQ ID NO:37 is the mutated version of the EPSPS sequence from rice that contains T173I, P177S and V403A mutations.

[0221] SEQ ID NO:38 is the mutated version of the EPSPS sequence from rice that contains T173I, P177T and V403A mutations.

[0222] SEQ ID NO:39 is the amino acid sequence of wild type EPSPS from CP4 (*Agrobacterium tumefaciens*).

1. Compositions

A. EPSPS Polypeptides

[0223] Various methods and compositions are provided which employ mutated polypeptides having EPSPS activity. Such EPSPS polypeptides comprise V403A mutation, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. Such EPSPS polypeptides comprise G172A and V403A mutation, wherein the amino acid positions correspond to the amino acid positions set forth in SEQ ID NO:1. Such EPSPS polypeptides comprise P177S and V403A mutation, wherein the amino acid positions correspond to the amino acid positions set forth in SEQ ID NO:1.

[0224] In some embodiments, the EPSPS polypeptides comprise G172A, P177S and/or V403A, wherein each amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0225] In some embodiments, the EPSPS polypeptides comprise T173I, P177S and/or V403A, wherein each amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0226] In some embodiments, the EPSPS polypeptides comprise T173I, P177T and/or V403A, wherein each amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0227] In other embodiments, the EPSPS polypeptides comprise G172A, P177S, V403A and at least one or more amino acid mutations selected from the group consisting of (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0228] In other embodiments, a plant EPSPS polypeptide comprises A70V, G172A, P177S, V403 and/or R514S mutations. In still other embodiments, a plant EPSPS polypeptide comprises A73T, G172A, P177S and/or V403A mutations. In still other embodiments, a plant EPSPS polypeptide comprises A73T, E86G, G172A, P177S and/or V403A mutations. In still other embodiments, a plant EPSPS polypeptide comprises A145G, G172A, P177S, V403A, T442S and/or D472E mutations. In still other embodiments, a plant EPSPS polypeptide comprises G172A, P177S, V403A and/or F502V mutations. In still other embodiments, a plant EPSPS polypeptide comprises G172A, P177S, Y252F and/or V403A mutations. In still other embodiments, a plant EPSPS polypeptide comprises E137D, G172A, P177S, L253M and/or V403A mutations. In still other embodiments, a plant EPSPS polypeptide comprises G172A, P177S, K399N, V403A and/or T466S mutations. In still other embodiments, a plant EPSPS polypeptide comprises A73G, L134F, G172A, P177S, G265V, S374R and/or V403A mutations. In still other embodiments, a plant EPSPS polypeptide comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A mutations. In still other embodiments, a plant EPSPS polypeptide comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A mutations. In still other embodiments, a plant EPSPS polypeptide comprises G172A, P177S, N246D and/or V403A mutations. In

still other embodiments, a plant EPSPS polypeptide comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I mutations. In still other embodiments, a plant EPSPS polypeptide comprises A73S, G172A, P177S, L214F and/or V403A mutations. In still other embodiments, a plant EPSPS polypeptide comprises G172A, P177S, V208A, L264H, V403A and/or R514K mutations. In still other embodiments, a plant EPSPS polypeptide comprises V113M, G172A, P177S and/or V403A mutations.

[0229] The EPSPS polypeptides mutants disclosed herein may have improved catalytic capacity in the presence of glyphosate when compared to wild type EPSPS polypeptides.

[0230] As disclosed herein, the EPSPS mutant comprising V403A can improve variety of plants resistant to glyphosate while still maintaining its own biological enzyme catalytic activity. However, the location of this amino acid V403 in the EPSPS may vary within other different plant species. The analogous mutation V->A on the analogous location may also improve other plants resistant to glyphosate. The mutation location can be decided in two steps. First step is to align the wild type plant EPSPS to rice EPSPS sequence (SEQ ID NO:1). Second step is to find the aligned position corresponding to the position 403 on the rice EPSPS sequence.

[0231] Previous study also discloses that G172 and P177 are also two conserved position on most of the EPSPS sequence. However, the location of this amino acid G172 or P177 in the EPSPS sequence may vary with some other species. The analogous mutation G->A, P->S and/or V->A on the analogous location may also improve some other plants resistant to glyphosate. The mutation location can be decided in two steps. First step is to align the wild type plant EPSPS to rice EPSPS sequence. Second step is to find the aligned position corresponding to the position 172, 177 and/or 403 on the rice EPSPS sequence.

[0232] Previous study also discloses that T173 and P177 are also two conserved position on most of the EPSPS sequence. However, the location of this amino acid T173 or P177 in the EPSPS sequence may vary with some other species. The analogous mutation T->I, P->S/T and/or V->A on the analogous location may also improve some other plants resistant to glyphosate. The mutation location can be decided in two steps. First step is to align the wild type plant EPSPS to rice EPSPS sequence. Second step is to find the aligned position corresponding to the position 173, 177 and/or 403 on the rice EPSPS sequence.

[0233] Various methods and compositions are provided which employ mutated polypeptides having EPSPS activity. Such EPSPS polypeptides comprise an amino acid mutation that is analogous to V403A mutation, wherein the amino acid position corresponds to the analogous amino acid position set forth in SEQ ID NO:1. Such EPSPS polypeptides comprise an amino acid mutation that is analogous to G172A and an amino acid mutation that is analogous to V403A mutation, wherein each amino acid position corresponds to the analogous amino acid position set forth in SEQ ID NO:1. Such EPSPS polypeptides comprise an amino acid mutation that is analogous to P177S and an amino acid mutation that is analogous to V403A mutation, wherein each amino acid position corresponds to the analogous amino acid position set forth in SEQ ID NO:1.

[0234] In some embodiments, the EPSPS polypeptides comprise an amino acid mutation that is analogous to

G172A, an amino acid mutation that is analogous to P177S and an amino acid mutation that is analogous to V403A, wherein each amino acid position corresponds to the analogous amino acid position set forth in SEQ ID NO: 1.

[0235] In some embodiments, the EPSPS polypeptides comprise an amino acid mutation that is analogous to T173I, an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A, wherein each amino acid position corresponds to the analogous amino acid position set forth in SEQ ID NO: 1.

[0236] In some embodiments, the EPSPS polypeptides comprise an amino acid mutation that is analogous to T173I, an amino acid mutation that is analogous to P177T and/or an amino acid mutation that is analogous to V403A, wherein each amino acid position corresponds to the analogous amino acid position set forth in SEQ ID NO: 1.

[0237] In other embodiments, the EPSPS polypeptides comprise an amino acid mutation that is analogous to G172A, an amino acid mutation that is analogous to P177S, an amino acid mutation that is analogous to V403A and at least one or more analogous amino acid mutations selected from the group consisting of (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO:1.

B. Plants and Other Host Cells of Interest

[0238] Further provided are engineered host cells that are transduced (transformed or transfected) with one or more EPSPS sequences or active variants or fragments thereof. The EPSPS polypeptides can be expressed in any organism, including non-animal cells such as plants, yeast, fungi, bacteria and the like.

[0239] Plants, plant cells, plant parts and seeds, and grain having the EPSPS sequences disclosed herein are also provided. In specific embodiments, the plants and/or plant parts have stably incorporated at least one heterologous EPSPS polypeptide disclosed herein or an active variant or fragment thereof. In addition, the plants or organism of interest can comprise multiple EPSPS polynucleotides.

[0240] In specific embodiments, the heterologous plant EPSPS polynucleotide in the plant or plant part is operably linked to a heterologous regulatory element, such as but not limited to a constitutive, tissue-preferred, or other promoter for expression in plants or a constitutive enhancer.

[0241] As used herein, the term "plant" includes plant cells, plant protoplasts, plant cell tissue cultures from which plants can be regenerated, plant calli, plant clumps, and plant cells that are intact in plants or parts of plants such as embryos, pollen, ovules, seeds, leaves, flowers, branches, fruit, kernels, ears, cobs, husks, stalks, roots, root tips, anthers, and the like. Grain is intended to mean the mature seed produced by commercial growers for purposes other than growing or reproducing the species. Progeny, variants, and mutants of the regenerated plants are also included within the scope of the disclosure, provided that these parts comprise the introduced polynucleotides.

[0242] The EPSPS sequences and active variants and fragments thereof disclosed herein may be used for transformation of any plant species, including, but not limited to, monocots and dicots. Examples of plant species of interest include, but are not limited to, corn (*Zea mays*), *Brassica* sp. (e.g., *B. napus*, *B. rapa*, *B. juncea*), particularly those *Brassica* species useful as sources of seed oil, barley (*Hordeum vulgare*), oat (*Avena sativa*), alfalfa (*Medicago sativa*), rice (*Oryza sativa*), rye (*Secale cereale*), sorghum (*Sorghum bicolor*, *Sorghum vulgare*), sweet sorghum (*Sorghum dochna*) millet (e.g., pearl millet (*Pennisetum glaucum*), proso millet (*Panicum miliaceum*), foxtail millet (*Setaria italica*), finger millet (*Eleusine coracana*)), hemp (*Cannabis sativa* L.), kentucky bluegrass (*Poa pratensis*), tall fescue (*Festuca arundinacea*), creeping bentgrass (*Agrostis stolonifera*), sunflower (*Helianthus annuus*), safflower (*Carthamus tinctorius*), wheat (*Triticum aestivum*), soybean (*Glycine max*), flax (*Linum usitatissimum*), tobacco (*Nicotiana tabacum*), potato (*Solanum tuberosum*), peanuts (*Arachis hypogaea*), cotton (*Gossypium barbadense*, *Gossypium hirsutum*), sweet potato (*Ipomoea batatas*), cassava (*Manihot esculenta*), coffee (*Coffea* spp.), coconut (*Cocos nucifera*), pineapple (*Ananas comosus*), citrus trees (*Citrus* spp.), cocoa (*Theobroma cacao*), tea (*Camellia sinensis*), banana (*Musa* spp.), avocado (*Persea americana*), fig (*Ficus casica*), guava (*Psidium guajava*), mango (*Mangifera indica*), tomato (*Solanum lycopersicum*), potato (*Solanum tuberosum*), apple tree (*Malus pumila*), grapevine (*Vitis vinifera* L.), cucumber (*C. sativus*), olive (*Olea europaea*), papaya (*Carica papaya*), cashew (*Anacardium occidentale*), macadamia (*Macadamia integrifolia*), almond (*Prunus amygdalus*), sugar beets (*Beta vulgaris*), sugarcane (*Saccharum* spp.), vegetables, ornamentals, conifers, turf grasses (including cool seasonal grasses and warm seasonal grasses).

[0243] Vegetables include tomatoes (*Lycopersicon esculentum*), lettuce (e.g., *Lactuca sativa*), green beans (*Phaseolus vulgaris*), lima beans (*Phaseolus limensis*), peas (*Lathyrus* spp.), potato (*Solanum tuberosum*), carrot (*Daucus carota* subsp. *Sativus*), onion (*Allium cepa*) and members of the genus *Cucumis* such as cucumber (*C. sativus*), cantaloupe (*C. cantalupensis*), and musk melon (*C. melo*). Ornamentals include azalea (*Rhododendron* spp.), hydrangea (*Macrophyllahydrangea*), hibiscus (*Hibiscus rosasanensis*), roses (*Rosa* spp.), tulips (*Tulipa* spp.), daffodils (*Narcissus* spp.), petunias (*Petunia hybrida*), carnation (*Dianthus caryophyllus*), poinsettia (*Euphorbia pulcherrima*), and chrysanthemum.

[0244] Other plants of interest include Poaceae species such as *Eleusine Indica*, *Aegliops tauschii* subse. *Tausch*, *Setaria Italica*, *Sorghum Halepense*, *Triticum Aestivum*, *Lolium Multiflorum*, *Oryza Meyeriana* var. *granulata*, *Brachypodium Distachyon*, *Bromus Tectorum*, *Lolium Rigidum*, *Panicum Miliaceum*, *Dichanthelium Oligosanthos*, *Hordeum Vulgare*, *Panicum Hali* etc. Other plants of interest include Asteraceae species such as *Erigeron Canadensis*, *Artemisia Annuu*, *Tanacetum Cinerariifolium* etc. Other plants of interest include Arecaceae species such as *Elaeis Guineensis*, *Phoenix Dactylifera* etc. Other plants of interest include Orchidaceae such as *Apostasia Shenzhenica*, *Dendrobium Catenatum*, *Phalaenopsis* etc. Other plants of interest include Rosaceae such as *Fragaria Ananassa*, *Rosa Chinensis* etc.

[0245] Other plants of interest include grain plants that provide seeds of interest, oil seed plants, and leguminous plants. Seeds of interest include grain seeds, such as corn, wheat, barley, rice, sorghum, rye, etc. Oil-seed plants include cotton, soybean, safflower, sunflower, *Brassica*, maize, alfalfa, palm, coconut, etc. Leguminous plants include beans and peas. Beans include guar, locust bean, fenugreek, soybean, garden beans, cowpea, mungbean, lima bean, fava bean, lentils, chickpea, etc.

[0246] Additional host cells of interest can be a eukaryotic cell, a protoplast, a tissue culture cell, prokaryotic cell, and/or a bacterial cell, such as *E. coli* and others.

[0247] For example, in some embodiments, glyphosate tolerant rice plants are provided, in which the glyphosate tolerant rice plants express an endogenous EPSPS polypeptide that comprises V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S and/or V403A, wherein each amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises T173I, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises T173I, P177T and/or V403A, wherein each amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, and/or V403A, and at least one amino acid mutations selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0248] Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises A70V, G172A, P177S, V403 and/or R514S mutation. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises A73T, G172A, P177S and/or V403A mutation. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises A73T, E86G, G172A, P177S and/or V403A mutation. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises A145G, G172A, P177S, V403A, T442S and/or D472E mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, V403A and/or F502V mutations. Further, the glyphosate tolerant rice plant can express an endogenous EPSPS polypeptide that comprises G172A, P177S, Y252F and/or V403A mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises E137D, G172A, P177S, L253M and/or V403A mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S,

K399N, V403A and/or T466S mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises A73G, L134F, G172A, P177S, G265V, S374R and/or V403A mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, N246D and/or V403A mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises A73S, G172A, P177S, L214F and/or V403A mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, V208A, L264H, V403A and/or R514K mutations. Further, the glyphosate tolerant rice plant may express an endogenous EPSPS polypeptide that comprises V113M, G172A, P177S and/or V403A mutations.

[0249] For example, in some embodiments, glyphosate tolerant flax plants are provided, in which the glyphosate tolerant flax plants express an endogenous EPSPS polypeptide that comprises V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, and/or V403A and at least one amino acid mutations selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0250] Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises A70V, G172A, P177S, V403 and/or R514S mutation. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises A73T, G172A, P177S and/or V403A mutation. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises A145G, G172A, P177S, V403A, T442S and/or D472E mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, V403A and/or F502V mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, Y252F and/or V403A mutations.

Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises E137D, G172A, P177S, L253M and/or V403A mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, K399N, V403A and/or T466S mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises A73G, L134F, G172A, P177S, G265V, S374R and/or V403A mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, N246D and/or V403A mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises A73S, G172A, P177S, L214F and/or V403A mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, V208A, L264H, V403A and/or R514K mutations. Further, the glyphosate tolerant flax plant may express an endogenous EPSPS polypeptide that comprises V113M, G172A, P177S and/or V403A mutations.

[0251] For example, in some embodiments, glyphosate tolerant wheat plants are provided, in which the glyphosate tolerant wheat plants express an endogenous EPSPS polypeptide that comprises V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, and/or V403A, and at least one amino acid mutations selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0252] Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises A70V, G172A, P177S, V403A and/or R514S mutation. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises A73T, G172A, P177S and/or V403A mutation. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises A73T, E86G, G172A, P177S and/or V403A mutation. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises A145G, G172A, P177S, V403A, T442S and/or D472E mutations. Further, the glyphosate

tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, V403A and/or F502V mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, Y252F and/or V403A mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises E137D, G172A, P177S, L253M and/or V403A mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, K399N, V403A and/or T466S mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises A73G, L134F, G172A, P177S, G265V, S374R and/or V403A mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, N246D and/or V403A mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises A73S, G172A, P177S, L214F and/or V403A mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, V208A, L264H, V403A and/or R514K mutations. Further, the glyphosate tolerant wheat plant may express an endogenous EPSPS polypeptide that comprises V113M, G172A, P177S and/or V403A mutations.

[0253] For example, in some embodiments, glyphosate tolerant soybean plants are provided, in which the glyphosate tolerant soybean plants express an endogenous EPSPS polypeptide that comprises V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO:1. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S and/or V403A, wherein the amino acid position corresponds to the amino acid position set forth in SEQ ID NO: 1. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, and/or V403A, and at least one amino acid mutations selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1.

[0254] Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises A70V, G172A, P177S, V403A and/or R514S mutation. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises A73T, G172A, P177S and/or V403A mutation. Further, the glyphosate tolerant soybean plant may express an endogenous

EPSPS polypeptide that comprises A73T, E86G, G172A, P177S and/or V403A mutation. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises A145G, G172A, P177S, V403A, T442S and/or D472E mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, V403A and/or F502V mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, Y252F and/or V403A mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises E137D, G172A, P177S, L253M and/or V403A mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, K399N, V403A and/or T466S mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises A73G, L134F, G172A, P177S, G265V, S374R and/or V403A mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, N246D and/or V403A mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises A73S, G172A, P177S, L214F and/or V403A mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises G172A, P177S, V208A, L264H, V403A and/or R514K mutations. Further, the glyphosate tolerant soybean plant may express an endogenous EPSPS polypeptide that comprises V113M, G172A, P177S and/or V403A mutations.

C. Stacking Other Traits of Interest

[0255] In one embodiment, the present invention relates to transgenic plants having the polynucleotide construct having a promotor operably linked to a mutated EPSPS as described above. In addition, other herbicide tolerant proteins may be inserted in the transgenic plants to produce multiple herbicide resistant plants.

[0256] As used herein, the term “stacked” includes having the multiple traits present in the same plant or organism of interest. A trait, as used herein, refers to the phenotype derived from a particular sequence or groups of sequences. In one embodiment, the molecular stack comprises at least one additional polynucleotide that also confers tolerance to at least one sequence that confers tolerance to glyphosate by the same and/or different mechanism and/or at least one additional polynucleotide that confers tolerance to a second herbicide.

[0257] The mechanism of glyphosate tolerance produced by the EPSPS sequences disclosed herein may be combined with other modes of herbicide resistance to provide host cells, plants, plant explants and plant cells that are tolerant to glyphosate and one or more other herbicides. For instance, the mechanism of glyphosate tolerance conferred

by EPSPS may be combined with other modes of glyphosate tolerance known in the art. In other embodiments, the plant or plant cell or plant part having the EPSPS sequence or an active variant or fragment thereof may be stacked with, for example, one or more sequences that confer tolerance to: an ALS inhibitor; an HPPD inhibitor; 2,4-D; Other phenoxy auxin herbicides; aryloxyphenoxypropionate herbals; dicamba; glufosinate herbicides; herbicides which target the protox enzyme.

[0258] The plant or plant cell or plant part having the EPSPS sequence thereof can also be combined with at least one other trait to produce plants that further comprise a variety of desired trait combinations. For instance, the plant or plant cell or plant part having the EPSPS sequence or an active variant or fragment thereof may be stacked with polynucleotides encoding polypeptides having pesticidal and/or insecticidal activity, or a plant or plant cell or plant part having the EPSPS sequence thereof may be combined with a plant disease resistance gene.

[0259] These stacked combinations can be created by any method including, but not limited to, breeding plants by any conventional methodology, or genetic transformation.

II. Methods of Use

A. Methods of Generating Glyphosate Tolerant Plants

[0260] The terms “glyphosate tolerance” and “glyphosate resistance” are used interchangeably herein.

I. Introducing

[0261] Various methods can be used to introduce a sequence of interest into a host cell, plant or plant part. The methods of the disclosure do not depend on a particular method for introducing a sequence into an organism or a plant or plant part, only that the polynucleotide or polypeptides gains access to the interior of at least one cell of the organism or the plant. Methods for introducing polynucleotide or polypeptide into various organisms, including plants, are known in the art including, but not limited to, stable transformation methods, transient transformation methods, and virus-mediated methods.

[0262] In specific embodiments, the EPSPS sequences thereof can be provided to a plant using a variety of transient transformation methods. Such transient transformation methods include, but are not limited to, the introduction of the EPSPS protein or active variants and fragments thereof directly into the plant. Such methods include, for example, microinjection or particle bombardment.

[0263] In other embodiments, the EPSPS polynucleotide disclosed herein or active variants thereof may be introduced into plants by contacting plants with a virus or viral nucleic acids. Generally, such methods involve incorporating a nucleotide construct of the disclosure within a DNA or RNA molecule.

[0264] Method are known in the art for the targeted insertion of a polynucleotide at a specific location in the plant genome. In one embodiment, the insertion of the polynucleotide at a desired genomic location is achieved using a site-specific recombination system.

[0265] The cells that have been transformed may be grown into plants in accordance with conventional ways. These plants may then be grown, and either pollinated with the same transformed strain or different strains, and the resulting

progeny having constitutive expression of the desired phenotypic characteristic identified. Two or more generations may be grown to ensure that expression of the desired phenotypic characteristic is stably maintained and inherited and then seeds harvested to ensure expression of the desired phenotypic characteristic has been achieved. In this manner, the present disclosure provides transformed seed (also referred to as transgenic seed) having a polynucleotide disclosed herein, for example, as part of an expression cassette, stably incorporated into their genome.

[0266] Transformed plant cells which are derived by plant transformation techniques, including those discussed above, can be cultured to regenerate a whole plant which possesses the transformed genotype to glyphosate or a glyphosate analog.

[0267] One of skill will recognize that after the expression cassette containing the EPSPS gene is incorporated in transgenic plants and confirmed to be operable, it can be introduced into other plants by sexual crossing. Any of a number of standard breeding techniques can be used, depending upon the species to be crossed.

[0268] In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise V403A, wherein the amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprising in its genome the recombinant DNA construct. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprising in its genome the recombinant DNA construct. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise T173I, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprising in its genome the recombinant DNA construct. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise T173I, P177T and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO:1 and generating a glyphosate tolerant plant that comprising in its genome the recombinant DNA construct. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise G172A, P177S, and/or V403A, and at least one amino acid mutations

selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1 and generating a glyphosate tolerant plant that comprising in its genome the recombinant DNA construct.

[0269] In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise A70V, G172A, P177S, V403 and/or R514S mutation. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise A73T, G172A, P177S and/or V403A mutation. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise A73T, E86G, G172A, P177S and/or V403A mutation. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise A145G, G172A, P177S, V403A, T442S and/or D472E mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise G172A, P177S, V403A and/or F502V mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise G172A, P177S, Y252F and/or V403A mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise E137D, G172A, P177S, L253M and/or V403A mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise G172A, P177S, K399N, V403A and/or T466S mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide

encodes a plant EPSPS polypeptide that comprise A73G, L134F, G172A, P177S, G265V, S374R and/or V403A mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise E163K, G172A, P177S, A216V, L253M, H389R and/or V403A mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise G172A, P177S, N246D and/or V403A mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise G172A, M175A, P177S, M258L, K356E, V403A and/or F512I mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise A73S, G172A, P177S, L214F and/or V403A mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise G172A, P177S, V208A, L264H, V403A and/or R514K mutations. In some embodiments, the methods comprise introducing by way of expressing in a regenerable plant cell a recombinant DNA construct comprising a polynucleotide operably linked to at least one regulatory sequence, wherein the polynucleotide encodes a plant EPSPS polypeptide that comprise V113M, G172A, P177S and/or V403A mutations.

II. Modifying

[0270] In general, methods to modify or alter the host genomic DNA are known and available to one skilled in the art. For example, a pre-existing or endogenous EPSPS sequence in a host plant can be modified or altered in a site-specific fashion using one or more site-specific engineering systems. This includes altering the host DNA sequence or a pre-existing transgenic sequence including regulatory elements, coding and non-coding sequences. These methods are also useful in targeting nucleic acids to pre-engineered target recognition sequences in the genome. As an example, the genetically modified cell or plant described herein, is generated using “custom” or engineered endonucleases such as meganucleases produced to modify plant genomes. Another site-directed engineering is through the use of zinc finger domain recognition coupled with the restriction properties of restriction enzyme. A transcription

activator-like (TAL) effector-DNA modifying enzyme (TALE or TALEN) is also used to engineer changes in plant genome.

[0271] Site-specific modification of plant genomes can also be performed using CRISPR system (clustered regularly interspaced short palindromic repeats). For example, researchers can use the CRISPR system to generate DNA double strand breaks (DSBs) at specific sites in the plant genome, and use the error-prone non-homologous end joining (NHEJ) mechanism to generate the disclosed mutations. The specific mutations can also be obtained through the precise base substitution under the guidance of an exogenous DNA donor (donor). Such base editing tools include but not limited to cytosine base editor (CBE) and/or adenine base editor (ABE). Or the prime editing system can also be used, the PBS (primer binding site) sequence in the pegRNA guides the mutations into the genome by using a reverse transcription template (RT template) containing the target editing sequence to generate these mutations.

[0272] An endogenous plant EPSPS gene in a plant cell may be modified to encode a glyphosate tolerant EPSPS protein that comprises V403A, the amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, P177S and/or V403A, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises T173I, P177S and/or V403A wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises T173I, P177T and/or V403A wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises G192A, P177S, and/or V403A, and at least one amino acid mutations selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the amino acid position set forth in SEQ ID NO: 1.

[0273] An endogenous plant EPSPS gene in a plant cell may be modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to V403A, wherein the amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO: 1. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises an amino acid mutation that is analogous to G172A, an amino acid mutation that is analogous to P177S and/or an amino acid mutation that is analogous to V403A wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO: 1. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate

tolerant EPSPS protein that comprises an amino acid mutation that is analogous to G172A, an amino acid mutation that is analogous to P177S, and/or an amino acid mutation that is analogous to V403A and at least one analogous amino acid mutations selected from the group consisting of: (1) A70V, (2) A73T/V/S/K/G, (3) V113M, (4) D114H, (5) L134F, (6) V136M, (7) E137D, (8) A145G, (9) K161E, (10) E163K, (11) M175A, (12) V208A, (13) L214F, (14) A216V, (15) Y252F, (16) L253M, (17) M258L, (18) A260T, (19) L264H, (20) G265V, (21) N316D, (22) K356E, (23) S374R, (24) H389R, (25) K399N, (26) T442S, (27) T466S, (28) D472E, (29) F502V, (30) F512I, and (31) R514S/K, wherein each amino acid mutation position corresponds to the analogous amino acid position set forth in SEQ ID NO: 1.

[0274] An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises A70V, G172A, P177S, V403 and/or R514S mutation. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises A73T, G172A, P177S and/or V403A mutation. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises A73T, E86G, G172A, P177S and/or V403A mutation. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises A145G, G172A, P177S, V403A, T442S and/or D472E mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, P177S, V403A and/or F502V mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, P177S, Y252F and/or V403A mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises E137D, G172A, P177S, L253M and/or V403A mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, P177S, K399N, V403A and/or T466S mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises A73G, L134F, G172A, P177S, G265V, S374R and/or V403A mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and/or V403A mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises E163K, G172A, P177S, A216V, L253M, H389R and/or V403A mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, P177S, N246D and/or V403A mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, M175A, P177S, M258L, K356E, V403A and/or F512I mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises A73S, G172A, P177S, L214F and/or V403A mutations. An endogenous plant EPSPS gene in a plant cell can be modified to encode a glyphosate tolerant EPSPS protein that comprises G172A, P177S, V208A, L264H, V403A and/or R514K mutations. An endogenous plant EPSPS gene in a plant cell can be

modified to encode a glyphosate tolerant EPSPS protein that comprises V113M, G172A, P177S and/or V403A mutations. [0275] The endogenous plant EPSPS gene can be modified by a CRISPR/Cas guide RNA-mediated system, a Zn-finger nuclease-mediated system, a meganuclease-mediated system, an oligonucleobase-mediated system, TALEN, or any gene modification system known to one of ordinary skill in the art. Moreover, for the purpose herein, an endogenous plant EPSPS gene includes coding DNA and genomic DNA within and surrounding the coding DNA, such as for example, the promoter, intron, and terminator sequences.

EXAMPLES

[0276] In the following Examples, it should be understood that these Examples, while indicating embodiments of the invention, are given by way of illustration only. From the above discussion and these Examples, one skilled in the art can make various changes and modifications of the invention to adapt it to various usages and conditions. Such modifications are also intended to fall within the scope of the appended embodiments.

Example 1: Alignment of Rice EPSPS and Other Plant EPSPS Sequences

[0277] Using rice (*Oryza sativa*) full length EPSPS (SEQ ID NO: 1) as a reference, we aligned rice EPSPS (SEQ ID NO: 2), flax EPSPS (SEQ ID NO:26), wheat EPSPS (SEQ ID NO: 31) and soybean EPSPS (SEQ ID NO:33). The alignment result was shown in FIG. 1. It show the described mutation positions and the corresponding position in each plant species. All disclosed mutation positions were marked according to SEQ ID NO:1.

Example 2: The Glyphosate Resistance Tests for Each Rice EPSPS Mutants on the Mutation Position of 403

[0278] The present inventors found V403 on rice EPSPS sequence plays a role in the sensitivity to glyphosate. The inventors performed saturation mutagenesis on the position 403 of the native rice EPSPS sequence to find out whether the mutation at this site can reduce the sensitivity of the rice EPSPS to glyphosate. Using NNK (where N represents adenine, thymine, guanine, and cytosine, and K represents thymine and thymine nucleotide) primers as the degenerate codon of the position to be mutated, the amino acid V on the position 403 of rice EPSPS polypeptide was mutated to the other 19 amino acids. The mutant PCR product was carried out using the construct pKED2 expressing the OsWt (SEQ ID NO:2) (FIG. 8) as the template, with V403 F61 as the forward primer and V403 R 61 as the reverse primer. The V403 F 61 primer sequence was AATGCCTGATNNKGC-CATGACCC. The V403 R 61 primer sequence was TTGTT-CATGTTGACATCAACAG.

[0279] The experiment steps and results were as following.

[0280] a) PCR system: According to the primer and template mixed PCR system corresponding to the target vector in the above table, the forward primer was 1.25 μl, the reverse primer was 1.25 μl, the template plasmid was 2 μl, the Q5 PCR master mix was 12.5 μl, and ddH₂O 18 μl.

[0281] b) PCR reaction condition:

Temperature	Time	Number of Cycles
98° C.	2 min	30 cycles
98° C.	15 s	
61° C.	30 s	
72° C.	2 min 30 s	
72° C.	2 min	

[0282] c) Took out 5 μl of the above PCR product and added it to the buffer for agarose gel electrophoresis. The amplification was successful if a PCR product with a size of about 4.5 kb is obtained.

[0283] d) Using KLD enzyme mixture from NEB company to remove the template from the PCR product, and then the product was connected. The connection system was: KLD buffer 5 μl, KLD MIX 1 μl, PCR product 1 μl, ddH₂O 3 μl. After 5 minute ligation at room temperature, the PCR product was then transformed into *E. coli*.

[0284] e) The ligation product was transformed into the EPSPS-deficient *E. coli* Dh5α competent, and the EPSPS-deficient *E. coli* strain was constructed by the well-known Red homologous recombination method. (Guangxing Bai, Zhiwei Sun, Ying Huang, et al. Knockout of *E. coli* ClpP gene using Red recombination system[J]. Chinese Journal of Biochemistry and Molecular Biology (1): 45-48.) Both *E. coli* transformation method and the detection methods are conventionally known in the art. After resuscitation, the cells were washed twice with M9 medium, and then the washed cells were spread on M9 medium containing 0.2 mM glyphosate and cultured for 48 h. Sequencing every grown single colony (sequencing and its analysis is done by Tsingke Biological Technology Co., Ltd.), only V403A mutants was found to be resistant to 0.2 mM glyphosate.

Example 3: The Comparison of the Glyphosate Resistance Among Rice EPSPS Mutants

[0285] Example 2 show the mutation V403A on rice EPSPS could affect the rice EPSPS sensitivity to the glyphosate. The inventor designed this experiment to test if any mutation on other position could also improve the mutated rice EPSPS sensitivity to the glyphosate.

[0286] The detailed experiment step and testing results were as following.

TABLE 2

The construct of Rice OsGA, OsPS, OsGAVA, OsPSVA, OsGAPS, OsGPV, OsTIPS, OsTIPT, OsTIPSPA and OsTIPTVA mutants.				
Target construct	Primer	Primer sequence	Template	Annealing Temperature
OsGA	G172A F	GGGAACGCTGcAACTGCAA	OsWt	66° C.
	G172A R	CAAGAAGAGTTGCACTTCC		

TABLE 2-continued

The construct of Rice OsGA, OsPS, OsGAVA, OsPSVA, OsGAPS, OsGPV, OsTIPS, OsTIPT, OsTIPSVA and OsTIPTVA mutants.				
Target construct	Primer	Primer sequence	Template	Annealing Temperature
OsPS	P177S F	TGCAATGCGACCATTGACA	OsWt	66° C.
	P177S R	GC GTTCCAGCGTTCCCAAG		
OsGAPS	GAPS F	atgcgatCATTGACAGCAGCCG	OsWt	66° C.
	GAPS R	TGA tgcagttgCAGCGTTCCCAAG AAGA		
OsGAVA	G172AF	GGGAACGCTGcAACTGCAA	Os VA	66° C.
	G172A R	TG CAAGAAGAGTTGCACTTCC		
OsPSVA	P177S F	TGCAATGCGACCATTGACA	Os VA	66° C.
	P177S R	GC GTTCCAGCGTTCCCAAG		
OsGPV	GAPS F	atgcgatCATTGACAGCAGCCG	Os VA	66° C.
	GAPS R	TGA tgcagttgCAGCGTTCCCAAG AAGA		
Os TIPS	Os PS F	tgcgatCATTGACAGCAGCCG	OsWt	66° C.
	66 Os TI R	TGA ttgcaaTCCAGCGTTCCCAAG GA		
Os TIPT	Os PT F	tgcgaaCATTGACAGCAGCCG	OsWt	66° C.
	66 Os TIR	TGA ttgcaaTCCAGCGTTCCCAAG GA		
Os TIPSVA	Os PS F	tgcgatCATTGACAGCAGCCG	Os VA	66° C.
	66 Os TI R	TGA ttgcaaTCCAGCGTTCCCAAG GA		
Os TIPTVA	Os PT F	tgcgaaCATTGACAGCAGCCG	Os VA	66° C.
	66 Os TI R	TGA ttgcaaTCCAGCGTTCCCAAG GA		

[0287] The detailed experiment steps as following:

i: Obtain and Keep the Desired Rice EPSPS Mutant

[0288] a) PCR system: According to the primer and template mixed PCR system corresponding to the target vector in the above table, the forward primer was 1.25 μ l, the reverse primer was 1.25 μ l, the template plasmid was 2 μ l, the Q5 PCR master mix was 12.5 μ l, and ddH₂O was 18 μ l.

Temperature	Time	The number of cycles
98° C.	2 min	30 cycles
98° C.	15 s	
*° C.	30 s	
72° C.	2 min 30 s	
72° C.	2 min	

[0289] b) PCR reaction conditions were as following: the annealing temperature was set according to the annealing temperature corresponding to the target carrier in the above table.

[0290] *Set the annealing temperature as above table provided

[0291] c) Took out 5 μ l of the obtained PCR product and added it to the loading buffer for agarose gel electrophoresis. A band with a size of about 4.5 kb was considered successful.

[0292] d) Used NEB's KLD enzyme mixture to remove the template and connected the PCR product. The connection system was 5 μ l KLD buffer, 1 μ l KLD MIX, 1 μ l PCR product, and ddH₂O 3 μ l. After ligating at room temperature for 5 min, transformed it into *E. coli*.

[0293] e) The ligation product was transformed into Dh5a competent *E. coli* lacking EPSPS. After resuscitation, the cells were washed twice with M9 medium, and then the washed cells were spread on M9 medium and cultured for 48 h. Sequenced the grown single colony (Tsingke Biological Technology Co., Ltd.), and selected the correct mutant to save for future use.

Pick the following mutants for testing:

[0294] OsWt represents the wild type rice EPSPS (SEQ ID NO:2). OsVA represents the mutated rice EPSPS with muta-

tion V403A(SEQ ID NO:3). OsGA represents the mutated rice EPSPS with mutation G172A. (SEQ ID NO:7) OsPS represents the mutated rice EPSPS with mutation P177S. (SEQ ID NO:8) OsGAPS represents the mutated rice EPSPS with mutation G172A and P177S. (SEQ ID NO:9) OsGAVA represents the mutated rice EPSPS with mutation G172A and V403A(SEQ ID NO:4). OsPSVA represents the mutated rice EPSPS with mutation P177S and V403A(SEQ ID NO:5). OsGPV represents the mutated rice EPSPS with mutation G172A, P177S and V403A(SEQ ID NO:6). OsTIPS represents the mutated rice EPSPS with mutation T173I and P177S. (SEQ ID NO: 35). OsTIPT represents the mutated rice EPSPS with mutation T173I and P177T. (SEQ ID NO:36). OsTIPSV represents the mutated rice EPSPS with mutation T173I, P177S and V403A. (SEQ ID NO:37). OsTIPTVA represents the mutated rice EPSPS with mutation T173I, P177T and V403A. (SEQ ID NO:38). CP4 represents the wild type EPSPS from *Agrobacterium tumefaciens*. (SEQ ID NO:39)

ii: The Glyphosate Resistance Comparison Among Rice EPSPS Mutants

[0295] Each product obtained from the above steps were transformed into *E. coli* lacking EPSPS. By detecting the growth of the transformed *Escherichia coli* on M9 medium containing different glyphosate concentrations (0 mM, 0.5 mM, 2 mM, 4 mM, 5 mM, 25 mM, 75 mM, 100 mM), the 1 mM glyphosate here was equivalent to 169 mg/ml glyphosate acid, the glyphosate resistance of rice EPSPS mutants was verified. The results were shown in FIG. 2.

[0296] On the medium containing 0 mM glyphosate, the strains expressing all other genes can grow normally except that the *E. coli* lacking EPSPS containing OsGAPS cannot grow. This result indicated that except OsGAPS mutants, all other rice EPSPS mutants can be expressed in *E. coli* to compensate the function of original EPSPS in *E. coli*.

[0297] The OsGA, OsPS and OsVA EPSPS mutants can grow on 0.5 mM glyphosate medium, but the OsWt cannot grow, indicating that either G172A, P177S or V403A single mutation can improve the glyphosate resistance of the rice EPSPS mutant. Here, V403A is a newly discovered mutation to improve EPSPS glyphosate tolerance.

[0298] FIG. 2 show that the OsGAVA can grow on 75 mM glyphosate medium or any medium with lesser than 75 mM glyphosate and the OsPSVA can grow on 4 mM glyphosate medium or any medium with lesser than 4 mM glyphosate. The V403A mutation combining with either G172A or P177S mutation can improve the glyphosate resistance of the rice EPSPS mutant.

[0299] FIG. 2 further show the OsGPV can grow on 100 mM glyphosate medium while the glyphosate resistance was comparable to the glyphosate resistance from CP4 EPSPS. The additional mutation of V403A can improve the glyphosate resistance of OsGAVA or OsGAPS.

[0300] FIG. 3 further show the OsTIPTVA can grow on 100 mM and OsTIPTVA can grow on 75 mM glyphosate medium. The additional mutation of V403 A can improve the glyphosate resistance of OsTIPS or OsTIPT.

Example 4: The Glyphosate Resistance for Flax EPSPS Mutants

i: Obtain the Flax EPSPS Mutant

[0301] The flax (*Linum usitatissimum*) EPSPS mutant sequence(SEQ ID NO: 27) was synthesized and cloned into

the pKED2 vector to obtain the vector LuVA. The vector LuWt was created after introducing A403V mutation by PCR.

[0302] The detailed experiment steps as following:

Target Vector	primer name	primer sequence	Temp-late	Annealing temperature
LuWt	Lu 403V_F 63	ATGCCAGACGt	LuVA	63° C.
		TGCGATGACT		
	Lu 403V_R 63	TTTGTTCATGT		
		TAACATCAACA		
		GC		

[0303] a) PCR system: According to the primer and template mixed PCR system corresponding to the target vector in the above table, the forward primer was 1.25 µl, the reverse primer was 1.25 µl, the template plasmid was 2 µl, the Q5 PCR master mix was 12.5 µl, and ddH₂O was 18 µl.

Temperature	Time	The number of cycles
98° C.	2 min	30 cycles
98° C.	15 s	
63° C.	30 s	
72° C.	2 min 30 s	
72° C.	2 min	

[0304] b) PCR reaction conditions were as follows: the annealing temperature was set according to the annealing temperature corresponding to the target carrier in the above table.

[0305] c) Took out 5 µl of the obtained PCR product and added it to the loading buffer for agarose gel electrophoresis. A band with a size of about 4.5 kb was considered successful.

[0306] d) Used NEB's KLD enzyme mixture to remove the template and then connected the PCR product. The connection system was 5 µl KLD buffer, 1 µl KLD MIX, 1 µl PCR product, and ddH₂O 3 µl. After ligating at room temperature for 5 min, transformed it into *E. coli*.

[0307] e) The ligation product was transformed into Dh5a competent *E. coli* lacking EPSPS. After resuscitation, the cells were washed twice with M9 medium, and then the washed cells were spread on M9 medium and cultured for 48 h. Sequenced the grown single colony (Tsingke Biological Technology Co., Ltd.), and selected the correct mutant to save for future use.

ii: Resistance Test of Flax EPSPS Mutant and Comparison with Wild Type

[0308] The resistance results were shown in FIG. 4: LuWt (SEQ ID NO:26) represents the wild-type flax EPSPS gene, and LuVA (SEQ ID NO:27) represents the V403A single-point mutant of flax EPSPS.

[0309] With the presence of 1 mM and 2 mM glyphosate, the growth status of LuVA was significantly better than that of LuWt, indicating that the V403A mutation can increase the glyphosate resistance of flax EPSPS.

Example 5: The Glyphosate Resistance for the Soybean EPSPS Mutant

i: Obtain the Desired Soybean EPSPS Mutant

[0310] According to the wild-type sequence information of soybean (*Glycine max*) EPSPS (SEQ ID NO:36), the gene

was synthesized and inserted into the vector pKED2 to obtain the vector GmWt. The V403 site of GmWt was mutated to A by PCR, and the vector GmVA expressing the soybean EPSPS mutant protein containing the V403A mutation was obtained. The specific experimental steps were as following:

Target Vector	Primer Name	Primer Sequence	Template	Annealing Temperature
GmVA	Gm EP	ATGCCAGATGcTGC	GmWt	63° C.
	VA F 63	CATGACT		
	Gm EP	CTTGTTTCATATTG		
	VAR 63	ACATCAATGCC		

[0311] a) PCR system: According to the primer and template mixed PCR system corresponding to the target vector in the above table, the forward primer was 1.25 μl, the reverse primer was 1.25 μl, the template plasmid was 2 μl, the Q5 PCR master mix was 12.5 μl, and ddH₂O was 18 μl.

Temperature	Time	The number of cycles
98° C.	2 min	30 cycles
98° C.	15 s	
63° C.	30 s	
72° C.	2 min 30 s	
72° C.	2 min	

[0312] b) PCR reaction conditions were as follows: the annealing temperature was set according to the annealing temperature corresponding to the target carrier in the above table.

[0313] c) Took out 5 μl of the obtained PCR product and added it to the loading buffer for agarose gel electrophoresis. A band with a size of about 4.5 kb was considered successful.

[0314] d) Used NEB’s KLD enzyme mixture to remove the template and connected the PCR product. The connection system was 5 μl KLD buffer, 1 μl KLD MIX, 1 μl PCR product, and ddH₂O 3 μl. After ligating at room temperature for 5 min, transformed it into *E. coli*.

[0315] e) The ligation product was transformed into Dh5a competent *E. coli* lacking EPSPS. After resuscitation, the cells were washed twice with M9 medium, and then the washed cells were spread on M9 medium and cultured for 48 h. Sequenced the grown single colony (Tsingke Biological Technology Co., Ltd.), and selected the correct mutant to save for future use.

ii: Resistance Test of Soybean EPSPS Mutant and Comparison with Wild Type

[0316] The resistance results were shown in FIG. 4: GmWt (SEQ ID NO:33) represent the wild-type soybean EPSPS gene, and GmVA(SEQ ID NO:34) represent the V403A single-point mutant of soybean EPSPS.

[0317] Under the pressure of 0.5 mM glyphosate, the growth status of GmVA was significantly higher than that of GmWt, indicating that the V403A mutation can increase the glyphosate resistance of soybean EPSPS.

Example 6: The Glyphosate Resistance for the Wheat EPSPS Mutant

i: Experimental Steps for Obtaining Wheat EPSPS Mutants

[0318] The gene was synthesized according to the V403A mutant sequence information (SEQ ID NO:32) of the EPSPS gene on chromosome 7 of the wheat A genome and inserted into the vector pKED2 to obtain the wheat (*Triticum vulgare*) EPSPS mutant expression vector TaVA containing the V403A mutation. The A403 site of TaVA was mutated back to V by PCR, and a TaWt vector expressing the EPSPS wild-type gene on chromosome 7 of wheat A genome was obtained.

[0319] The specific experimental steps were as following:

Target Vector	Primer Name	Primer Sequence	Template	Annealing Temperature
TaWt	Ta 403V	ATGCCAGATGtCGCG	TaVA	64° C.
	F 64	ATGACT		
	Ta 403	TTTGTTTCATGTTGAC		
	VR 64	ATCAACAGC		

[0320] a) PCR system: According to the primer and template mixed PCR system corresponding to the target vector in the above table, the forward primer was 1.25 μl, the reverse primer was 1.25 μl, the template plasmid was 2 μl, the Q5 PCR master mix was 12.5 μl, and ddH₂O was 18 μl.

Temperature	Time	The Number of Cycles
98° C.	2 min	30 cycles
98° C.	15 s	
64° C.	30 s	
72° C.	2 min 30 s	
72° C.	2 min	

[0321] b) PCR reaction conditions were as follows: the annealing temperature was set according to the annealing temperature corresponding to the target carrier in the above table.

[0322] c) Took out 5 μl of the obtained PCR product and added it to the loading buffer for agarose gel electrophoresis. A band with a size of about 4.5 kb was considered successful.

[0323] d) Used NEB’s KLD enzyme mixture to remove the template and then connected the PCR product. The connection system was 5 μl KLD buffer, 1 μl KLD MIX, 1 μl PCR product, and ddH₂O 3 μl. After ligating at room temperature for 5 min, transformed it into *E. coli*.

[0324] e) The ligation product was transformed into Dh5a competent *E. coli* lacking EPSPS. After resuscitation, the cells were washed twice with M9 medium, and then the washed cells were spread on M9 medium and cultured for 48 h. Sequenced the grown single colony (Tsingke Biological Technology Co., Ltd.), and selected the correct mutant to save for future use.

ii: Resistance Test of Wheat EPSPS Mutant and Comparison with Wild Type

[0325] The resistance results were shown in FIG. 4: TaWt (SEQ ID NO:31) represents the wild-type EPSPS gene on

chromosome 7 of the wheat A genome, and TaVA (SEQ ID NO:32) represents the V403A single-point mutant of wheat EPSPS.

[0326] Under the pressure of 0.2 mM glyphosate, the growth state of TaVA was significantly higher than that of TaWt, indicating that the V403A mutation can increase the glyphosate resistance of wheat EPSPS.

Example 7: Resistance Identification of Flax EPSPS Mutants LuGPV, LuTIPT, LuTIPTVA

[0327] The G172 position and P177 position of LuVA were mutated into A and S at the same time by PCR, and a 3-point mutant LuGPV was obtained.

[0328] The T173 and P177 positions of LuWT were simultaneously mutated to I and T by PCR, and a 3-point mutant LuTIPT was obtained.

[0329] The T173 and P177 positions of LuVA were simultaneously mutated to I and T by PCR, and a 3-point mutant LuTIPTVA was obtained.

[0330] The specific experimental steps were as following:

Target Vector	Primer	Primer Sequence	Template	Annealing Temperature
LuGPV	Lu 172A	atgcggttcgCTGACA	LuVA	63° C.
	177S_F 61	GCTGCTGTACAG		
	Lu 172A	agctgttgcAGCATT		
	177S_R 61	TCCAAGGAAAAGTTC		
LuTIPT VA	Lu PT	tgcgtaCGCTGACAG	LuVA	64° C.
	F 64	CTGCTGTAA		
	Lu TI	tagctaTTCCAGCAT		
	R 64	TTCCAAGGAAAAG		
LuTIPT	Lu PT	tgcgtaCGCTGACAG	LuWT	64° C.
	F 64	CTGCTGTAA		
	Lu TI	tagctaTTCCAGCAT		
	R 64	TTCCAAGGAAAAG		

[0331] a) PCR system: According to the primer and template mixed PCR system corresponding to the target vector in the above table, the forward primer was 1.25 μ l, the reverse primer was 1.25 μ l, the template plasmid was 2 μ l, the Q5 PCR master mix was 12.5 μ l, and ddH₂O was 18 μ l.

Temperature	Time	The Number of Cycles
98° C.	2 min	30 cycles
98° C.	15 s	
63° C./64° C.	30 s	
72° C.	2 min 30 s	
72° C.	2 min	

[0332] b) PCR reaction conditions were as follows: the annealing temperature was set according to the annealing temperature corresponding to the target carrier in the above table.

[0333] c) Took out 5 μ l of the obtained PCR product and added it to the loading buffer for agarose gel electrophoresis. A band with a size of about 4.5 kb was considered successful.

[0334] d) Used NEB's KLD enzyme mixture to remove the template and connected the PCR product. The connection system was 5 μ l KLD buffer, 1 μ l KLD MIX, 1 μ l PCR product, and ddH₂O 3 μ l. After ligating at room temperature for 5 min, transformed it into *E. coli*.

[0335] e) The ligation product was transformed into Dh5a competent *E. coli* lacking EPSPS. After resuscitation, the cells were washed twice with M9 medium, and then the washed cells were spread on M9 medium and cultured for 48 h. Sequenced the grown single colony (Tsingke Biological Technology Co., Ltd.), and selected the correct mutant to save for future use.

ii: Resistance Test of EPSPS Mutant of Flax and Comparison with Wild Type

[0336] The resistance results were shown in FIG. 5: LuWt (SEQ ID NO:26) represents the EPSPS2 gene of flax, LuGPV (SEQ ID NO:30) represents the flax G172A+P177S+V403A three-point mutant. The resistance results were shown in FIG. 6: LuTIPT (SEQ ID NO:28) was the flax T173I+P177S double-point mutants, LuTIPTVA (SEQ ID NO:29) was flax T173I+P177T+V403A three-point mutants.

[0337] The results showed that the GPV mutant of flax had strong glyphosate resistance. From the colony saturation under 100 mM glyphosate pressure, the resistance of LuGPV was higher than CP4. The resistance of LuTIPTVA was higher than that of LuTIPT, indicating that V403A can also significantly increase the resistance level of the mutant in the flax EPSPS gene.

Example 7: The Resistance Test for Other Rice Mutants

[0338] In order to further improve the glyphosate resistance and develop highly resistant rice EPSPS mutants that can meet commercial production requirements, we used OsGPV as a template to perform random mutations, with high-concentration glyphosate screening, and to identify EPSPS mutant that can improve the glyphosate resistance.

i. Obtain the Rice Mutants

1) Error-Prone PCR

[0339] We used OsGPV (SEQ ID NO:6) as a template to perform error-prone PCR. (With GeneMorph II Random Mutagenesis Kit, #200550, Agilent Technologies).

[0340] Reaction requirement:

10x Mutazyme II reaction buffer	5 μ l
dNTP mix (200 μ M each final)	1 μ l
EP OsF	0.5 μ l
EP OsR	0.5 μ l
Mutazyme II DNA polymerase (2.5 U/ μ l)	1 μ l
ddH ₂ O	X μ l
OsGPV	300 ng
Total	50 μ l

[0341] Primer sequence as following:

EP OsF: tgtgaattcatggcggcgaaggcggaggagatc

EP OsR: gatcctgcaggtcagttcctgacgaaagtgc

[0342] Procedure of PCR reaction:

Temperature	Time	The Number of Cycles
95° C.	2 min	40 cycles
95° C.	30 s	
60° C.	30 s	

-continued

Temperature	Time	The Number of Cycles
72° C.	90 s	
72° C.	10 min	

2) Performed PCR Product Recovery and Purification on the Obtained PCR Product

3) Recovered Product and Vector Backbone Digestion

[0343] a) The recovered product was digested with EcoRI and SbfI, and then ligated to the pKED2 vector after the same double digestion with T₄ DNA ligase to obtain the ligation product to obtain the vector carrying the rice EPSPS gene mutant.

[0344] b) Transformed the above vector into *Escherichia coli* DH5 α with EPSPS gene knocked out. The transformation steps were as follows: add the above ligation product to 50 μ l *E. coli* DH5 α competent cells with EPSPS gene knocked out, mix well and incubate on ice for 30 min, then after a heat shock at 42° C. for 90 s, immediately place it in an ice bath on ice for 2 min after taking it out. After the ice bath, add 500 μ l of LB liquid medium and resuscitate at 37° C. at 180 r/min for 45 min. The transformed *Escherichia coli* was a mutant library expressing rice EPSPS mutant genes obtained on the template basis of OsGPV.

4) Resistance screening

[0345] 5) Centrifuged the transformed product obtained above at 3000 rpm for 5 min, discarded the supernatant, then resuspended it in 1 ml M9 liquid medium, added 200 μ l M9 liquid medium to resuspend and inoculated it on an M9 plate containing 75 mM glyphosate. Incubated at 37 degrees for 24 hours.

[0346] Pick and saved the single colonies grown in the above screening plates, and conduct further resistance identification and sequencing

[0347] ii. Resistance test of rice mutants. In this example, the wild-type rice EPSPS gene OsWT, the most widely used commercial glyphosate-resistant gene CP4EPSPS, the rice mutant OsTIPS, which had the same mutation type with commercial glyphosate-resistant corn GA21, and the rice mutant OsGPV with the highly glyphosate-resistant rice EPSPS mutant gene obtained in this study were transformed as control, and the newly obtained rice EPSPS mutants were transformed as the experimental group, and the glyphosate resistance among the control and the mutants were compared. The monoclonal bacteria of the control group and the experimental group were respectively inoculated into a medium containing 0, 100 mM and 125 mM glyphosate, and the growth of colonies in each group was observed.

[0348] The order of Samples (from left to right):

[0349] 1st Row: 1-8, 1-13, 1-19, 1-24, 1-27, 1-37, 1-38

[0350] 2nd Row: 1-42, 1-68, 1-88, 1-93, 1-98, 1-125, 1-139

[0351] 3rd Row: 1-193, 1-150, 1-217, 1-246, 1-250, 1-265, 1-386

[0352] 4th Row: 1-411, 1-547, 1-552, 1-599, 1-614, 1-631, 1-130

[0353] 5th Row: 2-5, 2-43, 2-45, 2-46, 2-47, 2-63, 2-121

[0354] 6th Row: 2-114, 2-133, 3-8, 3-25, 3-27, 3-30, 3-32

[0355] 7th Row: 3-33, OsTIPS, 3-36, CP4, Os Wt, Os GPV

[0356] In the above results, under the pressure of 100 mM glyphosate, except for 3-32 and OsTIPS that did not grow, all other experimental and control samples were able to tolerate/resist the glyphosate, but the growth status of the colonies was inconsistent. Based on the colony growth status under 125 mM glyphosate pressure, the colony growth status of the 16 mutants (1-68 (SEQ ID NO:10), 1-98(SEQ ID NO:11), 1-139(SEQ ID NO:12), 1-246(SEQ ID NO:13), 1-386(SEQ ID NO:14), 1-411(SEQ ID NO:15), 1-547(SEQ ID NO:16), 1-552(SEQ ID NO:17), 1-599(SEQ ID NO:18), (SEQ ID NO:19), 2-5(SEQ ID NO:20), 2-46 (SEQ ID NO:21), 2-47(SEQ ID NO:22), (SEQ ID NO:23), 3-33(SEQ ID NO:24), 3-36(SEQ ID NO:25) were significantly higher than OsGPV in colony growth status, and some of the mutants were significantly more resistant than CP4, showing increased resistance. The colony growth status of the other 16 mutant showed decreased resistance than OsGPV. The other 12 mutants had no significant difference in resistance from OsGPV in FIG. 7.

Example 8: Transgenic Rice Plant's Resistance to Glyphosate Herbicide

[0357] The rice seed used in this example is RICE KENG XIANG DA008-169, which is publicly available.

[0358] 1. Obtaining the expression vector.

[0359] Nucleic acid sequence encoding OsGPV mutants (comprising G172A, P177S and V403A) (SEQ ID NO:6) were inserted into the plant expression vector according to the conventional methods to obtain the pk510 vector expressing the OsGPV gene. (see FIG. 9)

[0360] 2. Transgenic experiments in rice callus.

[0361] 1) The transgenic vector pk510 was introduced into *Agrobacterium* EHA105 to obtain recombinant *Agrobacterium*.

[0362] 2) One to five days before the transformation experiment, recombinant *Agrobacterium* was inoculated on YEB solid medium containing 50 mg/L kanamycin and 15 mg/L rifampicin. The medium was sealed in the dish with 3M tape; kept at temperature at 28° C. in an inverted dish orientation and cultured it in dark for one to five days.

[0363] 3) The bacteria was scraped at the "Z"-shaped tail with the inoculation ring, and gently suspended in 100 μ M NB liquid medium containing 2 mg/L 2,4-dichlorophenoxy-acetic acid to make the concentration of the bacteria liquid reach approximately O.D. 600=0.1, and then reserve the make *Agrobacterium* infection solution.

[0364] 4) After removing the rice seed coat, the rice seed was placed in a triangular flask, washed 3 times with sterile water. 75% alcohol was added to immerse the seeds, shaken gently for 1 min, and the alcohol discarded. Then, 2.5% sodium chlorate aqueous solution was added and the solution shaken at 150-170 rpm for 25 min. Next, sterile water was added and gently shake for 5-8 times, then pour out water. The seeds were inoculated on a callus induction medium and cultured in the dark at 30° C. for 4-6 weeks and rice callus obtained.

[0365] 5) The rice callus obtained in step 4 was soaked in the *Agrobacterium* infection solution for 10 minutes, and then placed on a co-culture medium covered with two layers of sterilized filter paper, and cultured in the dark at 22° C. for 3 days.

[0366] 6) The rice callus obtained in step 5 was taken and put it on the recovery medium, and cultivated for 4-7 days at 30° C.

[0367] 7) The rice callus obtained in step 6 was taken and placed on the hygromycin selection medium, and cultured at 30° C. for 2 weeks.

[0368] 8) The rice callus obtained in step 7 was taken and placed on the hygromycin selection medium, and cultured at 30° C. for 2 weeks.

[0369] 9) The vigorously growing callus was transferred to Differentiation medium ingredients and cultured at 30° C. for 20-30 days. Plants were produced from yellowish or green callus.

[0370] 10) Robust regenerated green seedlings larger than 3 cm in height were selected, transferred to rooting medium, and cultivated for 7-14 days to obtain rice TO seedlings.

Medium Ingredient:

[0371] Co-culture medium ingredients: NB solid medium containing 2 mg/L 2,4-dichlorophenoxyacetic acid

[0372] recovery medium ingredients: NB solid medium containing 200 mg/L timentin

[0373] hygromycin selection medium ingredients: NB solid medium containing 30-50 mg/L hygromycin

[0374] Differentiation medium ingredients: NB solid medium containing 0.5 g/L glutamine, 0.5 g/L proline, 2 mg/L kanamycin and 0.2 mg/L α -naphthoacetic acid.

[0375] Rooting medium ingredients: MS solid medium containing 0.5 g/L glutamine, 0.5 g/L proline, 0.2 mg/L α -naphthaleneacetic acid.

[0376] Transgenic rice TO seedlings were cultivated in an artificial climate chamber until harvest, and T1 generation rice seeds were obtained for the glyphosate resistance test.

[0377] 2. Identification of glyphosate resistance in T1 generation transgenic rice.

[0378] Transgenic rice TO seedlings were cultivated in an artificial climate chamber until harvest, and T1 generation rice seeds were obtained for the glyphosate resistance test.

[0379] 1) T1 generation seeds that were positive for transgene under ultraviolet light were selected. Since the plant expression vector was constructed by expressing OsGPV and the DsRed gene that can express red fluorescent protein in tandem, the T1 generation seeds can be differentiated by exposure to ultraviolet light. The seeds of red light were the seeds which contain both DsRed and OsGPV vectors.

[0380] 2) Accelerating germination: 20 seeds from each of the non-transgenic control seeds and T1 transgenic seeds containing the OsGPV gene were taken, and soaked in clean water for 4 hours to make the seeds fully absorb water. Then the seeds were transferred to a petri dish with two layers of filter paper at the bottom, and added a small amount of clean water to it to keep it moist. The amount of water should not overfill the seeds.

[0381] 3) After culturing for 2 days following the above method, the seeds were transferred to a petri dish (diameter 9 cm) with two layers of filter paper at the bottom, and a small amount of 5 ml of 100 ppm glyphosate aqueous solution was added (Roundup, adding 41% isopropylamine salt solution to dilute 10,000 times) and cultivated for 10 days.

[0382] 4) Ten rice seedlings were randomly picked and photos were taken, see FIG. 10.

[0383] In FIG. 10, it show that the growth of the wild-type rice as control was significantly inhibited in the 100 ppm glyphosate solution, and the root length was also significantly inhibited. In the subsequent observations, all the wild type died and the root rotted. The transgenic rice containing OsGPV mutants was not significantly affected by glyphosate herbicide and continued to grow, indicating that the rice plants with the OsGPV mutants was the glyphosate tolerate/resistant.

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	370		375		380
Asp Tyr Cys	Ile Ile Thr	Pro Pro Glu	Lys Leu Asn	Ile Thr Ala	Ile
	385		390		400
Asp Thr Tyr	Asp Asp His	Arg Met Ala	Met Ala Phe	Ser Leu Ala	Ala
	405		410		415
Cys Ala Asp	Val Pro Val	Thr Ile Arg	Asp Pro Gly	Cys Thr Arg	Lys
	420		425		430
Thr Phe Pro	Asn Tyr Phe	Asp Val Leu	Ser Thr Phe	Val Arg Asn	
	435		440		445

<210> SEQ ID NO 3

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 3

Met Ala Ala	Lys Ala Glu	Glu Ile Val	Leu Gln Pro	Ile Arg Glu	Ile
1	5		10		15
Ser Gly Ala	Val Gln Leu	Pro Gly Ser	Lys Ser Leu	Ser Asn Arg	Ile
	20		25		30
Leu Leu Leu	Ser Ala Leu	Ser Glu Gly	Thr Thr Val	Val Asp Asn	Leu
	35		40		45
Leu Asn Ser	Glu Asp Val	His Tyr Met	Leu Glu Ala	Leu Lys Ala	Leu
	50		55		60
Gly Leu Ser	Val Glu Ala	Asp Lys Val	Ala Lys Arg	Ala Val Val	Val
	65		70		75
Gly Cys Gly	Gly Lys Phe	Pro Val Glu	Lys Asp Ala	Lys Glu Glu	Val
	85		90		95
Gln Leu Phe	Leu Gly Asn	Gly Ala Thr	Ala Met Arg	Pro Leu Thr	Ala
	100		105		110
Ala Val Thr	Ala Ala Gly	Gly Asn Ala	Thr Tyr Val	Leu Asp Gly	Val
	115		120		125
Pro Arg Met	Arg Glu Arg	Pro Ile Gly	Asp Leu Val	Val Gly Leu	Lys
	130		135		140
Gln Leu Gly	Ala Asp Val	Asp Cys Phe	Leu Gly Thr	Glu Cys Pro	Pro
	145		150		155
Val Arg Val	Lys Gly Ile	Gly Gly Leu	Pro Gly Gly	Lys Val Lys	Leu
	165		170		175
Ser Gly Ser	Ile Ser Ser	Gln Tyr Leu	Ser Ala Leu	Leu Met Ala	Ala
	180		185		190
Pro Leu Ala	Leu Gly Asp	Val Glu Ile	Glu Ile Ile	Asp Lys Leu	Ile
	195		200		205
Ser Ile Pro	Tyr Val Glu	Met Thr Leu	Arg Leu Met	Glu Arg Phe	Gly
	210		215		220
Val Lys Ala	Glu His Ser	Asp Ser Trp	Asp Arg Phe	Tyr Ile Lys	Gly

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225		230		235		240									
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala
				245					250					255	
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr
			260					265					270		
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys
		275						280				285			
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp
	290					295					300				
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305					310					315					320
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala
			325						330					335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
		340						345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
	355							360				365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405						410					415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
		420						425					430		
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
	435							440				445			

<210> SEQ ID NO 4
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 4

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1			5						10					15	
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25						30		
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
	35						40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50					55					60				
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val
65				70					75					80	
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val
			85					90					95		
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Met	Arg	Pro	Leu	Thr	Ala
		100						105					110		
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val
	115						120					125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys

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130				135				140							
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro
145					150					155				160	
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu
			165						170					175	
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Met	Ala	Ala
			180						185					190	
Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile
		195					200						205		
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly
	210					215					220				
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly
225					230					235				240	
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala
			245						250					255	
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr
			260						265					270	
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys
		275					280						285		
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp
	290					295					300				
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305					310					315				320	
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala
			325						330					335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
			340						345					350	
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
	355						360						365		
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395				400	
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405						410					415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
		420							425				430		
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
	435						440						445		

<210> SEQ ID NO 5

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 5

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1			5						10				15		
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25					30			
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu

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35	40	45
Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu		
50	55	60
Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val		
65	70	75 80
Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val		
	85	90 95
Gln Leu Phe Leu Gly Asn Gly Ala Thr Ala Met Arg Ser Leu Thr Ala		
	100	105 110
Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val		
	115	120 125
Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys		
	130	135 140
Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro		
145	150	155 160
Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu		
	165	170 175
Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala		
	180	185 190
Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile		
	195	200 205
Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly		
	210	215 220
Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly		
225	230	235 240
Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala		
	245	250 255
Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr		
	260	265 270
Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys		
	275	280 285
Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp		
	290	295 300
Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys		
305	310	315 320
His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala		
	325	330 335
Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile		
	340	345 350
Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala		
	355	360 365
Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro		
	370	375 380
Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile		
385	390	395 400
Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala		
	405	410 415
Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys		
	420	425 430
Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn		
	435	440 445

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<210> SEQ ID NO 6
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 6

Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
1      5      10      15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
      20      25      30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
      35      40      45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
      50      55      60

Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
      65      70      75      80

Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
      85      90      95

Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala
      100     105     110

Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
      115     120     125

Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
      130     135     140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
      145     150     155     160

Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
      165     170     175

Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
      180     185     190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
      195     200     205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
      210     215     220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
      225     230     235     240

Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
      245     250     255

Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
      260     265     270

Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
      275     280     285

Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
      290     295     300

Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
      305     310     315     320

His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala
      325     330     335

Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
      340     345     350

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Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
   355                               360                   365

Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro
   370                               375                   380

Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile
   385                               390                   395                   400

Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala
                   405                               410                   415

Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys
                   420                               425                   430

Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn
                   435                               440                   445

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<210> SEQ ID NO 7
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.

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<400> SEQUENCE: 7

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Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
1      5      10      15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
20     25     30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
35     40     45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
50     55     60

Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
65     70     75     80

Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
85     90     95

Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Pro Leu Thr Ala
100    105    110

Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
115    120    125

Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
130    135    140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
145    150    155    160

Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
165    170    175

Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
180    185    190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
195    200    205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
210    215    220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
225    230    235    240

Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
245    250    255

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Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
    260                265                270

Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
    275                280                285

Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
    290                295                300

Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
    305                310                315                320

His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Val Ala
    325                330                335

Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
    340                345                350

Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
    355                360                365

Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro
    370                375                380

Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile
    385                390                395                400

Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala
    405                410                415

Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys
    420                425                430

Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn
    435                440                445

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<210> SEQ ID NO 8
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.

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<400> SEQUENCE: 8

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Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
 1          5          10          15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
 20          25          30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
 35          40          45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
 50          55          60

Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
 65          70          75          80

Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
 85          90          95

Gln Leu Phe Leu Gly Asn Ala Gly Thr Ala Met Arg Ser Leu Thr Ala
100          105          110

Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
115          120          125

Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
130          135          140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
145          150          155          160

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Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu
				165					170					175	
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Met	Ala	Ala
			180					185					190		
Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile
		195					200					205			
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly
	210					215					220				
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly
225					230					235					240
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala
				245					250					255	
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr
			260					265					270		
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys
		275					280					285			
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp
	290					295					300				
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305					310					315					320
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Val	Ala
				325					330					335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
		340						345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
		355					360					365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405					410					415		
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
		420						425					430		
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
	435						440					445			

<210> SEQ ID NO 9

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 9

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1			5						10				15		
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25						30		
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
		35					40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50					55					60				

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Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val	65	70	75	80
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val	85	90	95	
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Met	Arg	Ser	Leu	Thr	Ala	100	105	110	
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val	115	120	125	
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys	130	135	140	
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro	145	150	155	160
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu	165	170	175	
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Met	Ala	Ala	180	185	190	
Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile	195	200	205	
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly	210	215	220	
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly	225	230	235	240
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala	245	250	255	
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr	260	265	270	
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys	275	280	285	
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp	290	295	300	
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys	305	310	315	320
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Val	Ala	325	330	335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile	340	345	350	
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala	355	360	365	
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro	370	375	380	
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile	385	390	395	400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala	405	410	415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys	420	425	430	
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn		435	440	445	

<210> SEQ ID NO 10

<211> LENGTH: 447

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 10

Met Val Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
1          5          10          15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
20          25          30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
35          40          45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
50          55          60

Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
65          70          75          80

Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
85          90          95

Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala
100         105         110

Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
115         120         125

Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
130         135         140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
145         150         155         160

Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
165         170         175

Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
180         185         190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
195         200         205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
210         215         220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
225         230         235         240

Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
245         250         255

Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
260         265         270

Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
275         280         285

Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
290         295         300

Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
305         310         315         320

His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala
325         330         335

Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
340         345         350

Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
355         360         365

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Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro
 370          375          380

Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile
385          390          395          400

Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala
          405          410          415

Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys
          420          425          430

Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Ser Asn
          435          440          445

<210> SEQ ID NO 11
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 11

Met Ala Ala Lys Thr Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
 1          5          10          15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
          20          25          30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
          35          40          45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
          50          55          60

Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
          65          70          75          80

Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
          85          90          95

Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala
          100          105          110

Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
          115          120          125

Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
          130          135          140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
          145          150          155          160

Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
          165          170          175

Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
          180          185          190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
          195          200          205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
          210          215          220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
          225          230          235          240

Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
          245          250          255

Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
          260          265          270

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Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys
		275					280					285			
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp
	290					295					300				
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305					310					315					320
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala
			325						330					335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
			340					345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
		355					360					365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405					410						415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
			420					425					430		
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
		435					440					445			

<210> SEQ ID NO 12

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 12

Met	Ala	Ala	Lys	Thr	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1				5					10					15	
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25						30		
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
		35					40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50				55					60					
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val
65				70					75					80	
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Gly	Lys	Asp	Ala	Lys	Glu	Glu	Val
		85						90					95		
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Met	Arg	Ser	Leu	Thr	Ala
		100					105						110		
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val
		115				120						125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys
	130				135						140				
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro
145				150					155					160	
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu
			165				170							175	

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Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Met	Ala	Ala
			180					185					190		
Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile
		195					200					205			
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly
	210					215					220				
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly
225					230					235					240
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala
			245					250						255	
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr
			260					265					270		
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys
		275					280					285			
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp
	290					295					300				
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305					310					315					320
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala
			325					330						335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
			340					345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
		355					360					365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405					410					415		
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
			420					425				430			
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
		435					440					445			

<210> SEQ ID NO 13

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 13

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1			5					10					15		
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25					30			
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
		35				40					45				
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50				55					60					
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Gly	Val	Val	Val
65				70					75					80	

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Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val	
				85					90					95		
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Met	Arg	Ser	Leu	Thr	Ala	
				100					105					110		
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val	
				115					120					125		
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys	
				130					135					140		
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro	
				145					150					155		
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu	
				165					170					175		
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Met	Ala	Ala	
				180					185					190		
Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile	
				195					200					205		
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly	
				210					215					220		
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly	
				225					230					235		
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala	
				245					250					255		
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr	
				260					265					270		
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys	
				275					280					285		
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp	
				290					295					300		
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys	
				305					310					315		
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala	
				325					330					335		
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile	
				340					345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala	
				355					360					365		
Ile	Arg	Thr	Glu	Leu	Ser	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro	
				370					375					380		
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile	
				385					390					395		
Asp	Thr	Tyr	Glu	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala	
				405					410					415		
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys	
				420					425					430		
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn		
				435					440					445		

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<210> SEQ ID NO 14
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.
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<400> SEQUENCE: 14

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Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
1      5      10      15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
20     25     30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
35     40     45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
50     55     60

Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
65     70     75     80

Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
85     90     95

Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala
100    105    110

Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
115    120    125

Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
130    135    140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
145    150    155    160

Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
165    170    175

Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
180    185    190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
195    200    205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
210    215    220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
225    230    235    240

Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
245    250    255

Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
260    265    270

Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
275    280    285

Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
290    295    300

Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
305    310    315    320

His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala
325    330    335

Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
340    345    350

Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
355    360    365

Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro
370    375    380

Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile

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385	390	395	400
Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala	405	410	415
Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys	420	425	430
Thr Val Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn	435	440	445

<210> SEQ ID NO 15
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 15

Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile	1	5	10	15
Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile	20	25	30	
Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu	35	40	45	
Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu	50	55	60	
Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val	65	70	75	80
Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val	85	90	95	
Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala	100	105	110	
Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val	115	120	125	
Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys	130	135	140	
Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro	145	150	155	160
Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu	165	170	175	
Ser Gly Ser Ile Ser Ser Gln Phe Leu Ser Ala Leu Leu Met Ala Ala	180	185	190	
Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile	195	200	205	
Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly	210	215	220	
Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly	225	230	235	240
Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala	245	250	255	
Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr	260	265	270	
Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys	275	280	285	
Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp				

290					295					300					
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305					310					315				320	
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala
				325					330					335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
				340				345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
		355					360					365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
				405					410					415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
			420					425					430		
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
		435					440					445			
<210> SEQ ID NO 16															
<211> LENGTH: 447															
<212> TYPE: PRT															
<213> ORGANISM: Artificial Sequence															
<220> FEATURE:															
<223> OTHER INFORMATION: This sequence is synthesized in lab.															
<400> SEQUENCE: 16															
Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1				5					10					15	
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
			20					25					30		
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
		35					40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50					55					60				
Gly	Leu	Ser	Val	Asp	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val
65				70						75					80
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val
				85					90					95	
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Met	Arg	Ser	Leu	Thr	Ala
			100					105					110		
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val
		115					120					125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys
		130				135					140				
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro
145				150						155					160
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu
				165					170					175	
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Met	Ser	Ala	Leu	Leu	Met	Ala	Ala
				180				185					190		

-continued

195					200					205					
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly
210						215					220				
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly
225						230					235				240
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala
				245					250					255	
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr
			260					265					270		
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys
		275					280					285			
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp
290						295					300				
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305						310					315				320
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala
				325					330					335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
			340					345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
		355					360					365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
370						375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385						390					395				400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
				405					410					415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
		420						425					430		
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
		435					440					445			

<210> SEQ ID NO 17

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 17

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1			5						10					15	
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25						30		
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
		35					40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
		50				55					60				
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val
65				70						75				80	
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val
				85					90					95	
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Met	Arg	Ser	Leu	Thr	Ala

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100				105				110											
Ala	Val	Thr	115	Ala	Ala	Gly	120	Asn	Ala	Thr	125	Tyr	Val	Leu	125	Asp	Gly	Val	
Pro	Arg	Met	130	Arg	Glu	Arg	135	Pro	Ile	Gly	140	Asp	Leu	Val	140	Val	Gly	Leu	Lys
Gln	Leu	Gly	145	Ala	Asp	Val	150	Cys	Phe	Leu	155	Gly	Thr	Glu	155	Cys	Pro	Pro	160
Val	Arg	Val	165	Lys	Gly	Ile	165	Gly	Gly	Leu	170	Pro	Gly	Gly	170	Lys	Val	Lys	175
Ser	Gly	Ser	180	Ile	Ser	Ser	180	Gln	Tyr	Leu	185	Ser	Ala	Leu	185	Leu	Met	Ala	190
Pro	Leu	Ala	195	Leu	Gly	Asp	200	Val	Glu	Ile	200	Glu	Ile	Ile	205	Asp	Lys	Leu	Ile
Ser	Ile	Pro	210	Tyr	Val	Glu	215	Met	Thr	Leu	215	Arg	Leu	Met	220	Glu	Arg	Phe	Gly
Val	Lys	Ala	225	Glu	His	Ser	230	Asp	Ser	Trp	235	Asp	Arg	Phe	235	Tyr	Ile	Lys	240
Gly	Gln	Lys	245	Tyr	Lys	Ser	245	Pro	Gly	Asn	250	Ala	Tyr	Val	250	Glu	Gly	Asp	255
Ser	Ser	Ala	260	Ser	Tyr	Phe	260	Leu	Ala	Gly	265	Ala	Ala	Ile	265	Thr	Gly	Gly	270
Val	Thr	Val	275	Gln	Gly	Cys	280	Gly	Thr	Thr	280	Ser	Leu	Gln	285	Gly	Asp	Val	Lys
Phe	Ala	Glu	290	Val	Leu	Glu	295	Met	Met	Gly	295	Ala	Lys	Val	300	Thr	Trp	Thr	Asp
Thr	Ser	Val	305	Thr	Val	Thr	310	Gly	Pro	Pro	315	Arg	Glu	Pro	315	Tyr	Gly	Lys	320
His	Leu	Lys	325	Ala	Val	Asp	325	Val	Asn	Met	330	Asn	Asn	Met	330	Pro	Asp	Ala	335
Met	Thr	Leu	340	Ala	Val	Val	340	Ala	Leu	Phe	345	Ala	Asp	Gly	345	Pro	Thr	Ala	Ile
Arg	Asp	Val	355	Ala	Ser	Trp	360	Arg	Val	Lys	360	Glu	Thr	Glu	365	Arg	Met	Val	Ala
Ile	Arg	Thr	370	Glu	Leu	Thr	375	Lys	Leu	Gly	380	Ala	Ser	Val	380	Glu	Glu	Gly	Pro
Asp	Tyr	Cys	385	Ile	Ile	Thr	390	Pro	Pro	Glu	395	Lys	Leu	Asn	395	Ile	Thr	Ala	Ile
Asp	Thr	Tyr	405	Asp	Asp	His	405	Arg	Met	Ala	410	Met	Ala	Phe	410	Ser	Leu	Ala	415
Cys	Ala	Asp	420	Val	Pro	Val	425	Thr	Ile	Arg	425	Asp	Pro	Gly	430	Cys	Thr	Arg	Lys
Thr	Phe	Pro	435	Asn	Tyr	Phe	440	Asp	Val	Leu	440	Ser	Thr	Phe	445	Val	Arg	Asn	

<210> SEQ ID NO 18

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 18

Met Ala Ala Lys Gly Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile

-continued

1	5	10	15
Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile	20	25	30
Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu	35	40	45
Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu	50	55	60
Gly Phe Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val	65	70	75
Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val	85	90	95
Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala	100	105	110
Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val	115	120	125
Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys	130	135	140
Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro	145	150	155
Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu	165	170	175
Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala	180	185	190
Pro Leu Ala Leu Val Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile	195	200	205
Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly	210	215	220
Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly	225	230	235
Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala	245	250	255
Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr	260	265	270
Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys	275	280	285
Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp	290	295	300
Thr Arg Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys	305	310	315
His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala	325	330	335
Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile	340	345	350
Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala	355	360	365
Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro	370	375	380
Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile	385	390	395
Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala	405	410	415

-continued

Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys
420 425 430

Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn
435 440 445

<210> SEQ ID NO 19

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 19

Met Ala Ala Lys Lys Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
1 5 10 15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
20 25 30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val His Asn Leu
35 40 45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
50 55 60

Gly Leu Ser Met Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
65 70 75 80

Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Glu Glu Glu Val
85 90 95

Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala
100 105 110

Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
115 120 125

Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
130 135 140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
145 150 155 160

Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
165 170 175

Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Thr
180 185 190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
195 200 205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
210 215 220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
225 230 235 240

Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
245 250 255

Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
260 265 270

Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
275 280 285

Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
290 295 300

Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
305 310 315 320

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<210> SEQ ID NO 20
<211> LENGTH: 447
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.
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Met 1	Ala	Ala	Lys 5	Ala	Glu	Glu	Ile	Val 10	Leu	Gln	Pro	Ile	Arg	Glu 15	Ile
Ser	Gly	Ala	Val 20	Gln	Leu	Pro	Gly	Ser 25	Lys	Ser	Leu	Ser	Asn 30	Arg	Ile
Leu	Leu	Leu	Ser 35	Ala	Leu	Ser	Glu 40	Gly	Thr	Thr	Val	Val 45	Asp	Asn	Leu
Leu	Asn 50	Ser	Glu	Asp	Val	His 55	Tyr	Met	Leu	Glu	Ala 60	Leu	Lys	Ala	Leu
Gly 65	Leu	Ser	Val	Glu	Ala	Asp 70	Lys	Val	Ala	Lys 75	Arg	Ala	Val	Val	Val 80
Gly	Cys	Gly	Gly	Lys 85	Phe	Pro	Val	Glu	Lys 90	Asp	Ala	Lys	Glu	Lys 95	Val
Gln	Leu	Phe	Leu	Gly 100	Asn	Ala	Ala	Thr 105	Ala	Met	Arg	Ser	Leu	Thr	Ala
Ala	Val	Thr 115	Ala	Ala	Gly	Gly	Asn 120	Ala	Thr	Tyr	Val	Leu	Asp 125	Gly	Val
Pro	Arg 130	Met	Arg	Glu	Arg	Pro 135	Ile	Gly	Asp	Leu	Val 140	Val	Gly	Leu	Lys
Gln 145	Leu	Gly	Val	Asp 150	Val	Asp	Cys	Phe	Leu	Gly	Thr 155	Glu	Cys	Pro	Pro 160
Val	Arg	Val	Lys 165	Gly	Ile	Gly	Gly	Leu	Pro 170	Gly	Gly	Lys	Val	Lys 175	Leu
Ser	Gly	Ser	Ile 180	Ser	Ser	Gln	Tyr	Met 185	Ser	Ala	Leu	Leu	Met 190	Ala	Ala
Pro	Leu 195	Ala	Leu	Gly	Asp	Val	Glu 200	Ile	Glu	Ile	Ile 205	Asp	Lys	Leu	Ile
Ser	Ile 210	Pro	Tyr	Val	Glu	Met 215	Thr	Leu	Arg	Leu	Met 220	Glu	Arg	Phe	Gly

-continued

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
 225 230 235 240
 Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
 245 250 255
 Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
 260 265 270
 Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
 275 280 285
 Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
 290 295 300
 Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
 305 310 315 320
 Arg Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala
 325 330 335
 Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
 340 345 350
 Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
 355 360 365
 Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro
 370 375 380
 Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile
 385 390 395 400
 Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala
 405 410 415
 Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys
 420 425 430
 Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn
 435 440 445

<210> SEQ ID NO 21
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 21

Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
 1 5 10 15
 Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
 20 25 30
 Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
 35 40 45
 Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
 50 55 60
 Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
 65 70 75 80
 Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
 85 90 95
 Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala
 100 105 110
 Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
 115 120 125

-continued

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Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
 130          135          140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
145          150          155          160

Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
          165          170          175

Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
          180          185          190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
          195          200          205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
210          215          220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
225          230          235          240

Gly Gln Lys Tyr Lys Ser Pro Gly Asp Ala Tyr Val Glu Gly Asp Ala
          245          250          255

Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
          260          265          270

Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
          275          280          285

Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
          290          295          300

Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
305          310          315          320

His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala
          325          330          335

Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
          340          345          350

Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
          355          360          365

Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro
          370          375          380

Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile
385          390          395          400

Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala
          405          410          415

Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys
          420          425          430

Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn
          435          440          445

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<210> SEQ ID NO 22

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 22

```

Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
 1          5          10          15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
          20          25          30

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-continued

Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
	35						40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50					55					60				
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val
65				70						75					80
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val
				85					90					95	
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Ala	Arg	Ser	Leu	Thr	Ala
			100					105					110		
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val
		115					120					125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys
	130					135					140				
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro
145					150					155					160
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu
			165					170						175	
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Leu	Ala	Ala
			180					185					190		
Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile
	195						200					205			
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly
	210					215					220				
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly
225					230					235					240
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala
			245					250						255	
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr
		260						265					270		
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Glu
		275					280					285			
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp
	290					295					300				
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305					310					315					320
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala
			325					330						335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
		340						345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
		355					360					365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405					410						415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
		420						425						430	

-continued

Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Ile Val Arg Asn
 435 440 445

<210> SEQ ID NO 23
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 23

Met Ala Ala Lys Ser Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
 1 5 10 15
 Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
 20 25 30
 Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
 35 40 45
 Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
 50 55 60
 Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
 65 70 75 80
 Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
 85 90 95
 Gln Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala
 100 105 110
 Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
 115 120 125
 Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
 130 135 140
 Gln Phe Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
 145 150 155 160
 Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
 165 170 175
 Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
 180 185 190
 Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
 195 200 205
 Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
 210 215 220
 Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
 225 230 235 240
 Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
 245 250 255
 Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
 260 265 270
 Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
 275 280 285
 Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
 290 295 300
 Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
 305 310 315 320
 His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala
 325 330 335

-continued

Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
			340					345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
		355					360					365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405					410						415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
		420						425				430			
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
	435						440					445			

<210> SEQ ID NO 24

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 24

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1				5					10					15	
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25						30		
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
		35					40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50					55					60				
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val
65				70						75					80
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val
			85						90					95	
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Met	Arg	Ser	Leu	Thr	Ala
		100						105					110		
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val
		115				120						125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Ala	Val	Gly	Leu	Lys
	130					135					140				
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro
145				150						155				160	
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu
			165					170						175	
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Met	Ala	Ala
		180						185					190		
Pro	Leu	Ala	His	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile
		195				200						205			
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly
	210					215					220				
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly
225				230						235					240

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Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala	
				245					250					255		
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr	
			260					265					270			
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys	
		275					280					285				
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp	
	290					295					300					
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys	
305					310					315					320	
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala	
			325						330					335		
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile	
		340						345					350			
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala	
		355					360					365				
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro	
	370					375					380					
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile	
385					390					395					400	
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala	
			405					410						415		
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys	
		420						425					430			
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Lys	Asn		
		435					440					445				

<210> SEQ ID NO 25

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 25

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile	
1			5						10					15		
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile	
		20					25						30			
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Met	Asp	Asn	Leu	
		35					40					45				
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu	
	50					55				60						
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val	
65				70					75					80		
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val	
			85						90					95		
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Ala	Thr	Ala	Met	Arg	Ser	Leu	Thr	Ala	
		100						105					110			
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val	
		115					120						125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys	
	130					135					140					

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Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
 145 150 155 160
 Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
 165 170 175
 Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
 180 185 190
 Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
 195 200 205
 Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
 210 215 220
 Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
 225 230 235 240
 Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
 245 250 255
 Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
 260 265 270
 Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
 275 280 285
 Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
 290 295 300
 Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
 305 310 315 320
 His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala
 325 330 335
 Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
 340 345 350
 Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
 355 360 365
 Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro
 370 375 380
 Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile
 385 390 395 400
 Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala
 405 410 415
 Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys
 420 425 430
 Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn
 435 440 445

<210> SEQ ID NO 26
 <211> LENGTH: 445
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 26

Met Thr Val Pro Glu Glu Ile Val Leu Gln Pro Ile Lys Asp Ile Ser
 1 5 10 15
 Gly Ile Val Thr Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile Leu
 20 25 30
 Leu Leu Ala Ala Leu Ser Glu Gly Lys Thr Val Val Asp Asn Leu Leu
 35 40 45

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Asn	Ser	Asp	Asp	Val	His	Tyr	Met	Leu	Gly	Ala	Leu	Lys	Thr	Leu	Gly
50						55					60				
Leu	Asn	Val	Glu	His	Ser	Ser	Glu	Gln	Lys	Arg	Ala	Ile	Val	Glu	Gly
65					70					75				80	
Arg	Gly	Gly	Val	Phe	Pro	Val	Gly	Lys	Leu	Gly	Lys	Asn	Asp	Ile	Glu
			85					90						95	
Leu	Phe	Leu	Gly	Asn	Ala	Gly	Thr	Ala	Met	Arg	Pro	Leu	Thr	Ala	Ala
			100					105					110		
Val	Thr	Ala	Ala	Gly	Gly	Asn	Ser	Ser	Tyr	Ile	Leu	Asp	Gly	Val	Pro
		115					120					125			
Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys	Gln
	130					135					140				
Leu	Gly	Ala	Asp	Val	Ser	Cys	Ser	Ser	Thr	Ser	Cys	Pro	Pro	Val	His
145					150					155					160
Val	Asn	Ala	Lys	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu	Ser	Gly
			165					170						175	
Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Thr	Ala	Leu	Leu	Met	Ala	Ala	Pro	Leu
			180					185					190		
Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Val	Asp	Lys	Leu	Ile	Ser	Val
		195					200					205			
Pro	Tyr	Val	Asp	Met	Thr	Leu	Lys	Leu	Met	Glu	Arg	Phe	Gly	Val	Ala
	210					215					220				
Val	Glu	His	Ser	Gly	Ser	Trp	Asp	Arg	Phe	Phe	Val	Lys	Gly	Gly	Gln
225					230					235					240
Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala	Ser	Ser
			245						250					255	
Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr	Ile	Thr
			260				265						270		
Val	Glu	Gly	Cys	Gly	Thr	Ser	Ser	Leu	Gln	Gly	Asp	Val	Lys	Phe	Ala
		275					280					285			
Glu	Val	Leu	Glu	Lys	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Glu	Thr	Ser
	290					295					300				
Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Asp	Ala	Ser	Gly	Lys	Lys	His	Leu
305					310					315					320
Arg	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Val	Ala	Met	Thr
			325						330					335	
Leu	Ala	Val	Val	Ala	Leu	Tyr	Ala	Asp	Gly	Pro	Thr	Ala	Ile	Arg	Asp
			340				345						350		
Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Ile	Ala	Val	Cys
		355					360					365			
Thr	Glu	Leu	Arg	Lys	Leu	Gly	Ala	Thr	Val	Glu	Glu	Gly	Pro	Asp	Tyr
	370					375					380				
Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Ser	Ile	Ala	Glu	Ile	Asp	Thr
385					390					395					400
Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala	Cys	Ala
			405						410					415	
Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Lys	Lys	Thr	Phe
			420					425					430		
Pro	Asp	Tyr	Phe	Glu	Val	Leu	Glu	Arg	Tyr	Thr	Lys	His			
	435						440					445			

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<210> SEQ ID NO 27
<211> LENGTH: 445
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 27

Met Thr Val Pro Glu Glu Ile Val Leu Gln Pro Ile Lys Asp Ile Ser
1          5          10          15

Gly Ile Val Thr Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile Leu
20          25          30

Leu Leu Ala Ala Leu Ser Glu Gly Lys Thr Val Val Asp Asn Leu Leu
35          40          45

Asn Ser Asp Asp Val His Tyr Met Leu Gly Ala Leu Lys Thr Leu Gly
50          55          60

Leu Asn Val Glu His Ser Ser Glu Gln Lys Arg Ala Ile Val Glu Gly
65          70          75          80

Arg Gly Gly Val Phe Pro Val Gly Lys Leu Gly Lys Asn Asp Ile Glu
85          90          95

Leu Phe Leu Gly Asn Ala Gly Thr Ala Met Arg Pro Leu Thr Ala Ala
100         105         110

Val Thr Ala Ala Gly Gly Asn Ser Ser Tyr Ile Leu Asp Gly Val Pro
115         120         125

Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys Gln
130         135         140

Leu Gly Ala Asp Val Ser Cys Ser Ser Thr Ser Cys Pro Pro Val His
145         150         155         160

Val Asn Ala Lys Gly Gly Leu Pro Gly Gly Lys Val Lys Leu Ser Gly
165         170         175

Ser Ile Ser Ser Gln Tyr Leu Thr Ala Leu Leu Met Ala Ala Pro Leu
180         185         190

Ala Leu Gly Asp Val Glu Ile Glu Ile Val Asp Lys Leu Ile Ser Val
195         200         205

Pro Tyr Val Asp Met Thr Leu Lys Leu Met Glu Arg Phe Gly Val Ala
210         215         220

Val Glu His Ser Gly Ser Trp Asp Arg Phe Phe Val Lys Gly Gly Gln
225         230         235         240

Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala Ser Ser
245         250         255

Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr Ile Thr
260         265         270

Val Glu Gly Cys Gly Thr Ser Ser Leu Gln Gly Asp Val Lys Phe Ala
275         280         285

Glu Val Leu Glu Lys Met Gly Ala Lys Val Thr Trp Thr Glu Thr Ser
290         295         300

Val Thr Val Thr Gly Pro Pro Arg Asp Ala Ser Gly Lys Lys His Leu
305         310         315         320

Arg Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala Met Thr
325         330         335

Leu Ala Val Val Ala Leu Tyr Ala Asp Gly Pro Thr Ala Ile Arg Asp
340         345         350

Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Ile Ala Val Cys

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355	360	365
Thr Glu Leu Arg Lys Leu Gly Ala Thr Val Glu Glu Gly Pro Asp Tyr		
370	375	380
Cys Ile Ile Thr Pro Pro Glu Lys Leu Ser Ile Ala Glu Ile Asp Thr		
385	390	395 400
Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala Cys Ala		
405	410	415
Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Lys Lys Thr Phe		
420	425	430
Pro Asp Tyr Phe Glu Val Leu Glu Arg Tyr Thr Lys His		
435	440	445

<210> SEQ ID NO 28
 <211> LENGTH: 445
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 28

Met Thr Val Pro Glu Glu Ile Val Leu Gln Pro Ile Lys Asp Ile Ser
1 5 10 15
Gly Ile Val Thr Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile Leu
20 25 30
Leu Leu Ala Ala Leu Ser Glu Gly Lys Thr Val Val Asp Asn Leu Leu
35 40 45
Asn Ser Asp Asp Val His Tyr Met Leu Gly Ala Leu Lys Thr Leu Gly
50 55 60
Leu Asn Val Glu His Ser Ser Glu Gln Lys Arg Ala Ile Val Glu Gly
65 70 75 80
Arg Gly Gly Val Phe Pro Val Gly Lys Leu Gly Lys Asn Asp Ile Glu
85 90 95
Leu Phe Leu Gly Asn Ala Gly Ile Ala Met Arg Thr Leu Thr Ala Ala
100 105 110
Val Thr Ala Ala Gly Gly Asn Ser Ser Tyr Ile Leu Asp Gly Val Pro
115 120 125
Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys Gln
130 135 140
Leu Gly Ala Asp Val Ser Cys Ser Ser Thr Ser Cys Pro Pro Val His
145 150 155 160
Val Asn Ala Lys Gly Gly Leu Pro Gly Gly Lys Val Lys Leu Ser Gly
165 170 175
Ser Ile Ser Ser Gln Tyr Leu Thr Ala Leu Leu Met Ala Ala Pro Leu
180 185 190
Ala Leu Gly Asp Val Glu Ile Glu Ile Val Asp Lys Leu Ile Ser Val
195 200 205
Pro Tyr Val Asp Met Thr Leu Lys Leu Met Glu Arg Phe Gly Val Ala
210 215 220
Val Glu His Ser Gly Ser Trp Asp Arg Phe Phe Val Lys Gly Gly Gln
225 230 235 240
Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala Ser Ser
245 250 255
Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr Ile Thr

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260					265					270					
Val	Glu	Gly	Cys	Gly	Thr	Ser	Ser	Leu	Gln	Gly	Asp	Val	Lys	Phe	Ala
	275						280					285			
Glu	Val	Leu	Glu	Lys	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Glu	Thr	Ser
	290					295					300				
Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Asp	Ala	Ser	Gly	Lys	Lys	His	Leu
305					310					315					320
Arg	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Val	Ala	Met	Thr
			325						330						335
Leu	Ala	Val	Val	Ala	Leu	Tyr	Ala	Asp	Gly	Pro	Thr	Ala	Ile	Arg	Asp
			340					345					350		
Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Ile	Ala	Val	Cys
		355					360					365			
Thr	Glu	Leu	Arg	Lys	Leu	Gly	Ala	Thr	Val	Glu	Glu	Gly	Pro	Asp	Tyr
	370					375					380				
Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Ser	Ile	Ala	Glu	Ile	Asp	Thr
385					390					395					400
Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala	Cys	Ala
			405						410						415
Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Lys	Lys	Thr	Phe
			420					425					430		
Pro	Asp	Tyr	Phe	Glu	Val	Leu	Glu	Arg	Tyr	Thr	Lys	His			
	435						440					445			

<210> SEQ ID NO 29

<211> LENGTH: 445

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 29

Met	Thr	Val	Pro	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Lys	Asp	Ile	Ser
1				5					10					15	
Gly	Ile	Val	Thr	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile	Leu
		20						25					30		
Leu	Leu	Ala	Ala	Leu	Ser	Glu	Gly	Lys	Thr	Val	Val	Asp	Asn	Leu	Leu
		35					40					45			
Asn	Ser	Asp	Asp	Val	His	Tyr	Met	Leu	Gly	Ala	Leu	Lys	Thr	Leu	Gly
	50					55					60				
Leu	Asn	Val	Glu	His	Ser	Ser	Glu	Gln	Lys	Arg	Ala	Ile	Val	Glu	Gly
65				70						75				80	
Arg	Gly	Gly	Val	Phe	Pro	Val	Gly	Lys	Leu	Gly	Lys	Asn	Asp	Ile	Glu
			85					90						95	
Leu	Phe	Leu	Gly	Asn	Ala	Gly	Ile	Ala	Met	Arg	Thr	Leu	Thr	Ala	Ala
		100					105						110		
Val	Thr	Ala	Ala	Gly	Gly	Asn	Ser	Ser	Tyr	Ile	Leu	Asp	Gly	Val	Pro
		115					120					125			
Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys	Gln
	130					135					140				
Leu	Gly	Ala	Asp	Val	Ser	Cys	Ser	Ser	Thr	Ser	Cys	Pro	Pro	Val	His
145					150					155				160	
Val	Asn	Ala	Lys	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu	Ser	Gly

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      165      170      175
Ser Ile Ser Ser Gln Tyr Leu Thr Ala Leu Leu Met Ala Ala Pro Leu
      180      185      190

Ala Leu Gly Asp Val Glu Ile Glu Ile Val Asp Lys Leu Ile Ser Val
      195      200      205

Pro Tyr Val Asp Met Thr Leu Lys Leu Met Glu Arg Phe Gly Val Ala
      210      215      220

Val Glu His Ser Gly Ser Trp Asp Arg Phe Phe Val Lys Gly Gly Gln
      225      230      235      240

Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala Ser Ser
      245      250      255

Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr Ile Thr
      260      265      270

Val Glu Gly Cys Gly Thr Ser Ser Leu Gln Gly Asp Val Lys Phe Ala
      275      280      285

Glu Val Leu Glu Lys Met Gly Ala Lys Val Thr Trp Thr Glu Thr Ser
      290      295      300

Val Thr Val Thr Gly Pro Pro Arg Asp Ala Ser Gly Lys Lys His Leu
      305      310      315      320

Arg Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala Met Thr
      325      330      335

Leu Ala Val Val Ala Leu Tyr Ala Asp Gly Pro Thr Ala Ile Arg Asp
      340      345      350

Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Ile Ala Val Cys
      355      360      365

Thr Glu Leu Arg Lys Leu Gly Ala Thr Val Glu Glu Gly Pro Asp Tyr
      370      375      380

Cys Ile Ile Thr Pro Pro Glu Lys Leu Ser Ile Ala Glu Ile Asp Thr
      385      390      395      400

Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala Cys Ala
      405      410      415

Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Lys Lys Thr Phe
      420      425      430

Pro Asp Tyr Phe Glu Val Leu Glu Arg Tyr Thr Lys His
      435      440      445

<210> SEQ ID NO 30
<211> LENGTH: 445
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 30

Met Thr Val Pro Glu Glu Ile Val Leu Gln Pro Ile Lys Asp Ile Ser
1          5          10          15

Gly Ile Val Thr Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile Leu
      20          25          30

Leu Leu Ala Ala Leu Ser Glu Gly Lys Thr Val Val Asp Asn Leu Leu
      35          40          45

Asn Ser Asp Asp Val His Tyr Met Leu Gly Ala Leu Lys Thr Leu Gly
      50          55          60

Leu Asn Val Glu His Ser Ser Glu Gln Lys Arg Ala Ile Val Glu Gly

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65	70	75	80
Arg Gly Gly Val Phe Pro Val Gly Lys Leu Gly Lys Asn Asp Ile Glu	85	90	95
Leu Phe Leu Gly Asn Ala Ala Thr Ala Met Arg Ser Leu Thr Ala Ala	100	105	110
Val Thr Ala Ala Gly Gly Asn Ser Ser Tyr Ile Leu Asp Gly Val Pro	115	120	125
Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys Gln	130	135	140
Leu Gly Ala Asp Val Ser Cys Ser Ser Thr Ser Cys Pro Pro Val His	145	150	155
Val Asn Ala Lys Gly Gly Leu Pro Gly Gly Lys Val Lys Leu Ser Gly	165	170	175
Ser Ile Ser Ser Gln Tyr Leu Thr Ala Leu Leu Met Ala Ala Pro Leu	180	185	190
Ala Leu Gly Asp Val Glu Ile Glu Ile Val Asp Lys Leu Ile Ser Val	195	200	205
Pro Tyr Val Asp Met Thr Leu Lys Leu Met Glu Arg Phe Gly Val Ala	210	215	220
Val Glu His Ser Gly Ser Trp Asp Arg Phe Phe Val Lys Gly Gly Gln	225	230	235
Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala Ser Ser	245	250	255
Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr Ile Thr	260	265	270
Val Glu Gly Cys Gly Thr Ser Ser Leu Gln Gly Asp Val Lys Phe Ala	275	280	285
Glu Val Leu Glu Lys Met Gly Ala Lys Val Thr Trp Thr Glu Thr Ser	290	295	300
Val Thr Val Thr Gly Pro Pro Arg Asp Ala Ser Gly Lys Lys His Leu	305	310	315
Arg Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala Met Thr	325	330	335
Leu Ala Val Val Ala Leu Tyr Ala Asp Gly Pro Thr Ala Ile Arg Asp	340	345	350
Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Ile Ala Val Cys	355	360	365
Thr Glu Leu Arg Lys Leu Gly Ala Thr Val Glu Glu Gly Pro Asp Tyr	370	375	380
Cys Ile Ile Thr Pro Pro Glu Lys Leu Ser Ile Ala Glu Ile Asp Thr	385	390	395
Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala Cys Ala	405	410	415
Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Lys Lys Thr Phe	420	425	430
Pro Asp Tyr Phe Glu Val Leu Glu Arg Tyr Thr Lys His	435	440	445

<210> SEQ ID NO 31

<211> LENGTH: 446

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 31

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Met Ala Gly Ala Glu Glu Val Val Leu Gln Pro Ile Arg Glu Ile Ser
1      5      10      15

Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile Leu
20      25      30

Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu Leu
35      40      45

Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Glu Ala Leu Gly
50      55      60

Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val Gly
65      70      75      80

Cys Gly Gly Arg Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val Lys
85      90      95

Leu Phe Leu Gly Asn Ala Gly Thr Ala Met Arg Pro Leu Thr Ala Ala
100     105     110

Val Val Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val Pro
115     120     125

Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Gln Gln
130     135     140

Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Asn Cys Pro Pro Val
145     150     155     160

Arg Ile Asn Gly Lys Gly Gly Leu Pro Gly Gly Lys Val Lys Leu Ser
165     170     175

Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ser Leu Leu Met Ala Ala Pro
180     185     190

Leu Ala Leu Glu Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile Ser
195     200     205

Val Pro Tyr Val Glu Met Thr Leu Lys Leu Met Glu His Phe Gly Val
210     215     220

Thr Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly Gly
225     230     235     240

Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala Ser
245     250     255

Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr Val
260     265     270

Thr Val Glu Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys Phe
275     280     285

Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp Thr
290     295     300

Ser Val Thr Val Thr Gly Pro Pro Arg Gln Pro Phe Gly Arg Lys His
305     310     315     320

Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Val Ala Met
325     330     335

Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile Arg
340     345     350

Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala Ile
355     360     365

Arg Thr Glu Leu Thr Lys Leu Gly Ala Thr Val Glu Glu Gly Pro Asp
370     375     380

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Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile Asp
385 390 395 400

Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala Cys
405 410 415

Ala Glu Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys Thr
420 425 430

Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Lys Asn
435 440 445

<210> SEQ ID NO 32

<211> LENGTH: 446

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 32

Met Ala Gly Ala Glu Glu Val Val Leu Gln Pro Ile Arg Glu Ile Ser
1 5 10 15

Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile Leu
20 25 30

Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu Leu
35 40 45

Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Glu Ala Leu Gly
50 55 60

Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val Gly
65 70 75 80

Cys Gly Gly Arg Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val Lys
85 90 95

Leu Phe Leu Gly Asn Ala Gly Thr Ala Met Arg Pro Leu Thr Ala Ala
100 105 110

Val Val Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val Pro
115 120 125

Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Gln Gln
130 135 140

Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Asn Cys Pro Pro Val
145 150 155 160

Arg Ile Asn Gly Lys Gly Gly Leu Pro Gly Gly Lys Val Lys Leu Ser
165 170 175

Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ser Leu Leu Met Ala Ala Pro
180 185 190

Leu Ala Leu Glu Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile Ser
195 200 205

Val Pro Tyr Val Glu Met Thr Leu Lys Leu Met Glu His Phe Gly Val
210 215 220

Thr Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly Gly
225 230 235 240

Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala Ser
245 250 255

Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr Val
260 265 270

Thr Val Glu Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys Phe
275 280 285

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Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp Thr
 290 295 300

Ser Val Thr Val Thr Gly Pro Pro Arg Gln Pro Phe Gly Arg Lys His
 305 310 315 320

Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala Met
 325 330 335

Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile Arg
 340 345 350

Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala Ile
 355 360 365

Arg Thr Glu Leu Thr Lys Leu Gly Ala Thr Val Glu Glu Gly Pro Asp
 370 375 380

Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile Asp
 385 390 395 400

Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala Cys
 405 410 415

Ala Glu Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys Thr
 420 425 430

Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Lys Asn
 435 440 445

<210> SEQ ID NO 33
 <211> LENGTH: 452
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 33

Met Ala Ala Ala Glu Lys Pro Ser Thr Ala Pro Glu Ile Val Leu Glu
 1 5 10 15

Pro Ile Lys Asp Ile Ser Gly Thr Ile Thr Leu Pro Gly Ser Lys Ser
 20 25 30

Leu Ser Asn Arg Ile Leu Leu Leu Ala Ala Leu Ser Glu Gly Thr Thr
 35 40 45

Val Val Asp Asn Leu Leu Tyr Ser Glu Asp Ile His Tyr Met Leu Gly
 50 55 60

Ala Leu Arg Thr Leu Gly Leu Arg Val Glu Asp Gln Thr Thr Lys
 65 70 75 80

Gln Ala Ile Val Glu Gly Cys Gly Gly Leu Phe Pro Thr Ile Lys Glu
 85 90 95

Ser Lys Asp Glu Ile Asn Leu Phe Leu Gly Asn Ala Gly Thr Ala Met
 100 105 110

Arg Pro Leu Thr Ala Ala Val Val Ala Ala Gly Gly Asn Ala Ser Tyr
 115 120 125

Val Leu Asp Gly Val Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu
 130 135 140

Val Ala Gly Leu Lys Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly
 145 150 155 160

Thr Asn Cys Pro Pro Val Arg Val Asn Gly Lys Gly Gly Leu Pro Gly
 165 170 175

Gly Lys Val Lys Leu Ser Gly Ser Ile Ser Ser Gln Tyr Leu Thr Ala
 180 185 190

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Leu Leu Met Ala Ala Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile
  195                200                205

Val Asp Lys Leu Ile Ser Val Pro Tyr Val Glu Met Thr Leu Lys Leu
  210                215                220

Met Glu Arg Phe Gly Val Ser Val Glu His Ser Gly Asn Trp Asp Lys
  225                230                235                240

Phe Leu Val His Gly Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Phe
                245                250                255

Val Glu Gly Asp Ala Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala
  260                265                270

Val Thr Gly Gly Thr Ile Thr Val Asn Gly Cys Gly Thr Asn Ser Leu
  275                280                285

Gln Gly Asp Val Lys Phe Ala Glu Val Leu Glu Lys Met Gly Ala Lys
  290                295                300

Val Thr Trp Ser Glu Asn Ser Val Thr Val Thr Gly Pro Pro Gln Asp
  305                310                315                320

Ser Ser Gly Gln Lys Val Leu Gln Gly Ile Asp Val Asn Met Asn Lys
                325                330                335

Met Pro Asp Val Ala Met Thr Leu Ala Val Val Ala Leu Phe Ala Asn
  340                345                350

Gly Gln Thr Ala Ile Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr
  355                360                365

Glu Arg Met Ile Ala Ile Cys Thr Glu Leu Arg Lys Leu Gly Ala Thr
  370                375                380

Val Glu Glu Gly Pro Asp Tyr Cys Val Ile Thr Pro Pro Glu Lys Leu
  385                390                395                400

Asn Val Thr Ala Ile Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala
                405                410                415

Phe Ser Leu Ala Ala Cys Gly Asp Val Pro Val Thr Ile Lys Asp Pro
  420                425                430

Gly Cys Thr Arg Lys Thr Phe Pro Asp Tyr Phe Glu Val Leu Glu Arg
  435                440                445

Phe Thr Arg His
  450

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<210> SEQ ID NO 34

<211> LENGTH: 452

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 34

```

Met Ala Ala Ala Glu Lys Pro Ser Thr Ala Pro Glu Ile Val Leu Glu
  1          5          10          15

Pro Ile Lys Asp Ile Ser Gly Thr Ile Thr Leu Pro Gly Ser Lys Ser
  20          25          30

Leu Ser Asn Arg Ile Leu Leu Leu Ala Ala Leu Ser Glu Gly Thr Thr
  35          40          45

Val Val Asp Asn Leu Leu Tyr Ser Glu Asp Ile His Tyr Met Leu Gly
  50          55          60

Ala Leu Arg Thr Leu Gly Leu Arg Val Glu Asp Asp Gln Thr Thr Lys
  65          70          75          80

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Gln	Ala	Ile	Val	Glu	Gly	Cys	Gly	Gly	Leu	Phe	Pro	Thr	Ile	Lys	Glu	
				85					90							
Ser	Lys	Asp	Glu	Ile	Asn	Leu	Phe	Leu	Gly	Asn	Ala	Gly	Thr	Ala	Met	
				100					105							
Arg	Pro	Leu	Thr	Ala	Ala	Val	Val	Ala	Ala	Gly	Gly	Asn	Ala	Ser	Tyr	
				115					120							
Val	Leu	Asp	Gly	Val	Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	
				130					135							
Val	Ala	Gly	Leu	Lys	Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	
				145					150							
Thr	Asn	Cys	Pro	Pro	Val	Arg	Val	Asn	Gly	Lys	Gly	Gly	Leu	Pro	Gly	
				165					170							
Gly	Lys	Val	Lys	Leu	Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Thr	Ala	
				180					185							
Leu	Leu	Met	Ala	Ala	Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	
				195					200							
Val	Asp	Lys	Leu	Ile	Ser	Val	Pro	Tyr	Val	Glu	Met	Thr	Leu	Lys	Leu	
				210					215							
Met	Glu	Arg	Phe	Gly	Val	Ser	Val	Glu	His	Ser	Gly	Asn	Trp	Asp	Lys	
				225					230							
Phe	Leu	Val	His	Gly	Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Phe	
				245					250							
Val	Glu	Gly	Asp	Ala	Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	
				260					265							
Val	Thr	Gly	Gly	Thr	Ile	Thr	Val	Asn	Gly	Cys	Gly	Thr	Asn	Ser	Leu	
				275					280							
Gln	Gly	Asp	Val	Lys	Phe	Ala	Glu	Val	Leu	Glu	Lys	Met	Gly	Ala	Lys	
				290					295							
Val	Thr	Trp	Ser	Glu	Asn	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Gln	Asp	
				305					310							
Ser	Ser	Gly	Gln	Lys	Val	Leu	Gln	Gly	Ile	Asp	Val	Asn	Met	Asn	Lys	
				325					330							
Met	Pro	Asp	Ala	Ala	Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asn	
				340					345							
Gly	Gln	Thr	Ala	Ile	Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	
				355					360							
Glu	Arg	Met	Ile	Ala	Ile	Cys	Thr	Glu	Leu	Arg	Lys	Leu	Gly	Ala	Thr	
				370					375							
Val	Glu	Glu	Gly	Pro	Asp	Tyr	Cys	Val	Ile	Thr	Pro	Pro	Glu	Lys	Leu	
				385					390							
Asn	Val	Thr	Ala	Ile	Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	
				405					410							
Phe	Ser	Leu	Ala	Ala	Cys	Gly	Asp	Val	Pro	Val	Thr	Ile	Lys	Asp	Pro	
				420					425							
Gly	Cys	Thr	Arg	Lys	Thr	Phe	Pro	Asp	Tyr	Phe	Glu	Val	Leu	Glu	Arg	
				435					440							
Phe	Thr	Arg	His													
				450												

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<210> SEQ ID NO 35
<211> LENGTH: 447
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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 35

Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
1          5          10          15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
          20          25          30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
          35          40          45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
          50          55          60

Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
          65          70          75          80

Gly Cys Gly Gly Lys Phe Pro Val Glu Lys Asp Ala Lys Glu Glu Val
          85          90          95

Gln Leu Phe Leu Gly Asn Ala Gly Ile Ala Met Arg Ser Leu Thr Ala
          100          105          110

Ala Val Thr Ala Ala Gly Gly Asn Ala Thr Tyr Val Leu Asp Gly Val
          115          120          125

Pro Arg Met Arg Glu Arg Pro Ile Gly Asp Leu Val Val Gly Leu Lys
          130          135          140

Gln Leu Gly Ala Asp Val Asp Cys Phe Leu Gly Thr Glu Cys Pro Pro
          145          150          155          160

Val Arg Val Lys Gly Ile Gly Gly Leu Pro Gly Gly Lys Val Lys Leu
          165          170          175

Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
          180          185          190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
          195          200          205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
          210          215          220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
          225          230          235          240

Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
          245          250          255

Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
          260          265          270

Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
          275          280          285

Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
          290          295          300

Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
          305          310          315          320

His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Val Ala
          325          330          335

Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
          340          345          350

Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
          355          360          365

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Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
370						375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405					410						415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
		420						425				430			
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
	435						440					445			

<210> SEQ ID NO 36
 <211> LENGTH: 447
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 36

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1			5						10					15	
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25						30		
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
	35						40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50					55					60				
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val
65				70						75				80	
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val
		85						90					95		
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Gly	Ile	Ala	Met	Arg	Thr	Leu	Thr	Ala
		100					105						110		
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val
	115					120						125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys
	130					135					140				
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro
145				150						155				160	
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu
		165						170					175		
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Met	Ala	Ala
		180					185						190		
Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile
	195					200						205			
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly
	210					215					220				
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly
225				230						235				240	
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala
		245						250						255	
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr
	260						265						270		

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Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys
		275					280					285			
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp
	290					295					300				
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys
305					310					315					320
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Val	Ala
			325						330					335	
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile
			340					345					350		
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala
		355					360					365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro
	370					375					380				
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile
385					390					395					400
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala
			405					410						415	
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys
			420					425					430		
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn	
		435					440					445			

<210> SEQ ID NO 37

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 37

Met	Ala	Ala	Lys	Ala	Glu	Glu	Ile	Val	Leu	Gln	Pro	Ile	Arg	Glu	Ile
1				5					10					15	
Ser	Gly	Ala	Val	Gln	Leu	Pro	Gly	Ser	Lys	Ser	Leu	Ser	Asn	Arg	Ile
		20					25						30		
Leu	Leu	Leu	Ser	Ala	Leu	Ser	Glu	Gly	Thr	Thr	Val	Val	Asp	Asn	Leu
		35					40					45			
Leu	Asn	Ser	Glu	Asp	Val	His	Tyr	Met	Leu	Glu	Ala	Leu	Lys	Ala	Leu
	50					55					60				
Gly	Leu	Ser	Val	Glu	Ala	Asp	Lys	Val	Ala	Lys	Arg	Ala	Val	Val	Val
65				70					75					80	
Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val
			85					90					95		
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Gly	Ile	Ala	Met	Arg	Ser	Leu	Thr	Ala
			100				105						110		
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val
		115					120					125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys
	130					135					140				
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro
145				150						155				160	
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu
			165					170						175	

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Ser Gly Ser Ile Ser Ser Gln Tyr Leu Ser Ala Leu Leu Met Ala Ala
    180              185              190

Pro Leu Ala Leu Gly Asp Val Glu Ile Glu Ile Ile Asp Lys Leu Ile
    195              200              205

Ser Ile Pro Tyr Val Glu Met Thr Leu Arg Leu Met Glu Arg Phe Gly
    210              215              220

Val Lys Ala Glu His Ser Asp Ser Trp Asp Arg Phe Tyr Ile Lys Gly
    225              230              235              240

Gly Gln Lys Tyr Lys Ser Pro Gly Asn Ala Tyr Val Glu Gly Asp Ala
    245              250              255

Ser Ser Ala Ser Tyr Phe Leu Ala Gly Ala Ala Ile Thr Gly Gly Thr
    260              265              270

Val Thr Val Gln Gly Cys Gly Thr Thr Ser Leu Gln Gly Asp Val Lys
    275              280              285

Phe Ala Glu Val Leu Glu Met Met Gly Ala Lys Val Thr Trp Thr Asp
    290              295              300

Thr Ser Val Thr Val Thr Gly Pro Pro Arg Glu Pro Tyr Gly Lys Lys
    305              310              315              320

His Leu Lys Ala Val Asp Val Asn Met Asn Lys Met Pro Asp Ala Ala
    325              330              335

Met Thr Leu Ala Val Val Ala Leu Phe Ala Asp Gly Pro Thr Ala Ile
    340              345              350

Arg Asp Val Ala Ser Trp Arg Val Lys Glu Thr Glu Arg Met Val Ala
    355              360              365

Ile Arg Thr Glu Leu Thr Lys Leu Gly Ala Ser Val Glu Glu Gly Pro
    370              375              380

Asp Tyr Cys Ile Ile Thr Pro Pro Glu Lys Leu Asn Ile Thr Ala Ile
    385              390              395              400

Asp Thr Tyr Asp Asp His Arg Met Ala Met Ala Phe Ser Leu Ala Ala
    405              410              415

Cys Ala Asp Val Pro Val Thr Ile Arg Asp Pro Gly Cys Thr Arg Lys
    420              425              430

Thr Phe Pro Asn Tyr Phe Asp Val Leu Ser Thr Phe Val Arg Asn
    435              440              445

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<210> SEQ ID NO 38

<211> LENGTH: 447

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: This sequence is synthesized in lab.

<400> SEQUENCE: 38

```

Met Ala Ala Lys Ala Glu Glu Ile Val Leu Gln Pro Ile Arg Glu Ile
1              5              10              15

Ser Gly Ala Val Gln Leu Pro Gly Ser Lys Ser Leu Ser Asn Arg Ile
    20              25              30

Leu Leu Leu Ser Ala Leu Ser Glu Gly Thr Thr Val Val Asp Asn Leu
    35              40              45

Leu Asn Ser Glu Asp Val His Tyr Met Leu Glu Ala Leu Lys Ala Leu
    50              55              60

Gly Leu Ser Val Glu Ala Asp Lys Val Ala Lys Arg Ala Val Val Val
    65              70              75              80

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Gly	Cys	Gly	Gly	Lys	Phe	Pro	Val	Glu	Lys	Asp	Ala	Lys	Glu	Glu	Val	
				85					90				95			
Gln	Leu	Phe	Leu	Gly	Asn	Ala	Gly	Ile	Ala	Met	Arg	Thr	Leu	Thr	Ala	
				100					105				110			
Ala	Val	Thr	Ala	Ala	Gly	Gly	Asn	Ala	Thr	Tyr	Val	Leu	Asp	Gly	Val	
				115					120				125			
Pro	Arg	Met	Arg	Glu	Arg	Pro	Ile	Gly	Asp	Leu	Val	Val	Gly	Leu	Lys	
				130					135				140			
Gln	Leu	Gly	Ala	Asp	Val	Asp	Cys	Phe	Leu	Gly	Thr	Glu	Cys	Pro	Pro	
				145					150				155			
Val	Arg	Val	Lys	Gly	Ile	Gly	Gly	Leu	Pro	Gly	Gly	Lys	Val	Lys	Leu	
				165					170				175			
Ser	Gly	Ser	Ile	Ser	Ser	Gln	Tyr	Leu	Ser	Ala	Leu	Leu	Met	Ala	Ala	
				180					185				190			
Pro	Leu	Ala	Leu	Gly	Asp	Val	Glu	Ile	Glu	Ile	Ile	Asp	Lys	Leu	Ile	
				195					200				205			
Ser	Ile	Pro	Tyr	Val	Glu	Met	Thr	Leu	Arg	Leu	Met	Glu	Arg	Phe	Gly	
				210					215				220			
Val	Lys	Ala	Glu	His	Ser	Asp	Ser	Trp	Asp	Arg	Phe	Tyr	Ile	Lys	Gly	
				225					230				235			
Gly	Gln	Lys	Tyr	Lys	Ser	Pro	Gly	Asn	Ala	Tyr	Val	Glu	Gly	Asp	Ala	
				245					250				255			
Ser	Ser	Ala	Ser	Tyr	Phe	Leu	Ala	Gly	Ala	Ala	Ile	Thr	Gly	Gly	Thr	
				260					265				270			
Val	Thr	Val	Gln	Gly	Cys	Gly	Thr	Thr	Ser	Leu	Gln	Gly	Asp	Val	Lys	
				275					280				285			
Phe	Ala	Glu	Val	Leu	Glu	Met	Met	Gly	Ala	Lys	Val	Thr	Trp	Thr	Asp	
				290					295				300			
Thr	Ser	Val	Thr	Val	Thr	Gly	Pro	Pro	Arg	Glu	Pro	Tyr	Gly	Lys	Lys	
				305					310				315			
His	Leu	Lys	Ala	Val	Asp	Val	Asn	Met	Asn	Lys	Met	Pro	Asp	Ala	Ala	
				325					330				335			
Met	Thr	Leu	Ala	Val	Val	Ala	Leu	Phe	Ala	Asp	Gly	Pro	Thr	Ala	Ile	
				340					345				350			
Arg	Asp	Val	Ala	Ser	Trp	Arg	Val	Lys	Glu	Thr	Glu	Arg	Met	Val	Ala	
				355					360				365			
Ile	Arg	Thr	Glu	Leu	Thr	Lys	Leu	Gly	Ala	Ser	Val	Glu	Glu	Gly	Pro	
				370					375				380			
Asp	Tyr	Cys	Ile	Ile	Thr	Pro	Pro	Glu	Lys	Leu	Asn	Ile	Thr	Ala	Ile	
				385					390				395			
Asp	Thr	Tyr	Asp	Asp	His	Arg	Met	Ala	Met	Ala	Phe	Ser	Leu	Ala	Ala	
				405					410				415			
Cys	Ala	Asp	Val	Pro	Val	Thr	Ile	Arg	Asp	Pro	Gly	Cys	Thr	Arg	Lys	
				420					425				430			
Thr	Phe	Pro	Asn	Tyr	Phe	Asp	Val	Leu	Ser	Thr	Phe	Val	Arg	Asn		
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<211> LENGTH: 459
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: This sequence is synthesized in lab.
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<400> SEQUENCE: 39

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Arg Lys Ser Ser Gly Leu Ser Gly Thr Val Arg Ile Pro Gly Asp Lys
20      25      30
Ser Ile Ser His Arg Ser Phe Met Phe Gly Gly Leu Ala Ser Gly Glu
35      40      45
Thr Arg Ile Thr Gly Leu Leu Glu Gly Glu Asp Val Ile Asn Thr Gly
50      55      60
Lys Ala Met Gln Ala Met Gly Ala Arg Ile Arg Lys Glu Gly Asp Thr
65      70      75      80
Trp Ile Ile Asp Gly Val Gly Asn Gly Gly Leu Leu Ala Pro Glu Ala
85      90      95
Pro Leu Asp Phe Gly Asn Ala Ala Thr Gly Cys Arg Leu Thr Met Gly
100     105     110
Leu Val Gly Val Tyr Asp Phe Asp Ser Thr Phe Ile Gly Asp Ala Ser
115     120     125
Leu Thr Lys Arg Pro Met Gly Arg Val Leu Asn Pro Leu Arg Glu Met
130     135     140
Gly Val Gln Val Lys Ser Glu Asp Gly Asp Arg Leu Pro Val Thr Leu
145     150     155     160
Arg Gly Pro Lys Thr Pro Thr Pro Ile Thr Tyr Arg Val Pro Met Ala
165     170     175
Ser Ala Gln Val Lys Ser Ala Val Leu Leu Ala Gly Leu Asn Thr Pro
180     185     190
Gly Ile Thr Thr Val Ile Glu Pro Ile Met Thr Arg Asp His Thr Glu
195     200     205
Lys Met Leu Gln Gly Phe Gly Ala Asn Leu Thr Val Glu Thr Asp Ala
210     215     220
Asp Gly Val Arg Thr Ile Arg Leu Glu Gly Arg Gly Lys Leu Thr Gly
225     230     235     240
Gln Val Ile Asp Val Pro Gly Asp Pro Ser Ser Thr Ala Phe Pro Leu
245     250     255
Val Ala Ala Leu Leu Val Pro Gly Ser Asp Val Thr Ile Leu Asn Val
260     265     270
Leu Met Asn Pro Thr Arg Thr Gly Leu Ile Leu Thr Leu Gln Glu Met
275     280     285
Gly Ala Asp Ile Glu Val Ile Asn Pro Arg Leu Ala Gly Gly Glu Asp
290     295     300
Val Ala Asp Leu Arg Val Arg Ser Ser Thr Leu Lys Gly Val Thr Val
305     310     315     320
Pro Glu Asp Arg Ala Pro Ser Met Ile Asp Glu Tyr Pro Ile Leu Ala
325     330     335
Val Ala Ala Ala Phe Ala Glu Gly Ala Thr Val Met Asn Gly Leu Glu
340     345     350
Glu Leu Arg Val Lys Glu Ser Asp Arg Leu Ser Ala Val Ala Asn Gly
355     360     365
Leu Lys Leu Asn Gly Val Asp Cys Asp Glu Gly Glu Thr Ser Leu Val
370     375     380
Val Arg Gly Arg Pro Asp Gly Lys Gly Leu Gly Asn Ala Ser Gly Ala

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385	390	395	400
Ala Val Ala Thr His Leu Asp His Arg Ile Ala Met Ser Phe Leu Val			
	405	410	415
Met Gly Leu Val Ser Glu Asn Pro Val Thr Val Asp Asp Ala Thr Met			
	420	425	430
Ile Ala Thr Ser Phe Pro Glu Phe Met Asp Leu Met Ala Gly Leu Gly			
	435	440	445
Ala Lys Ile Glu Leu Ser Asp Thr Lys Ala Ala			
450	455		

1. A polynucleotide encoding a plant EPSP synthase (EPSPS) polypeptide, wherein the plant EPSPS polypeptide comprises amino acid V mutating to amino acid A, wherein the mutation location is the location of the amino acid V in the EPSPS amino acid sequence of the plant corresponding to the location of V403 listed in SEQ ID NO:1.

2. (canceled)

3. The polynucleotide of claim 1, wherein the plant EPSPS polypeptide comprises G172A and V403A amino acid mutations,

wherein the plant EPSPS polypeptide comprises P177S and V403A amino acid mutations,

wherein the plant EPSPS polypeptide comprises G172A, P177S, and V403A amino acid mutations,

wherein the plant EPSPS polypeptide comprises T173I, P177S, and V403A amino acid mutations,

wherein the plant EPSPS polypeptide comprises T173I, P177T, and V403A amino acid mutations,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises A70V, G172A, P177S, V403A, and R514S,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises A73T, G172A, P177S, and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises A73T, E86G, G172A, P177S, and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises A145G, G172A, P177S, V403A, T442S, and D472E,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A, P177S, V403A and F502V,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A, P177S, Y252F and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises E137D, G172A, P177S, L253M and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A, P177S, K399N, V403A and T466S,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises A73G, L134G, G172A, P177S, G265V, S374R and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises A73K, D114H, V136M, K161E, G172A, P177S, A260T and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises E163K, G172A, P177S, A216V, L253M, H389R and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A, P177S, N247D and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A, M175A, P177S, M258L, K356E, V403A and F512I,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises A73S, G172A, P177S, L214F and V403A,

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises G172A, P177S, V208A, L264H, V403A and R514K, or

wherein the polynucleotide encodes a plant EPSPS polypeptide that comprises V113M, G172A, P177S and V403A.

4. The polynucleotide of claim 3, wherein the plant EPSPS polypeptide is shown as any one of SEQ ID NO: 3-25, 27-30, 32, and 34-38.

5. The polynucleotide of claim 4, wherein the plant EPSPS polypeptide is shown as SEQ ID NO: 6.

6-24. (canceled)

25. A recombinant DNA construct comprising the polynucleotide of claim 1.

26-27. (canceled)

28. A plant comprising in its genome the polynucleotide of claim 1.

29. The plant of claim 28, wherein said plant is corn, rice, wheat, barley, oat, *sorghum*, millet, flax, hemp, kentucky bluegrass, tall fescue, creeping bentgrass, *Brassica* sp., sugar beets, soybean, peas, alfalfa, tobacco, cotton, sunflower, tomato, potato, apple tree, grapevine, citrus trees, cucumber or *petunia*.

30. The plant of claim 29, wherein said plant is rice.

31-107. (canceled)

* * * * *